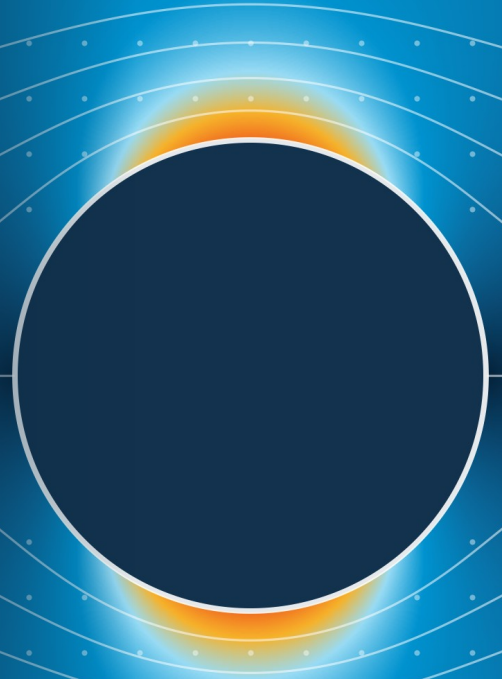
AIChE
KAUST Student Chapter

KAUST AIChE STUDENT CHAPTER · WORKSHOP

COMSOL Multiphysics Workshop

From CFD fundamentals to hands-on simulation



Thursday, 29 October 2026

1:00 – 3:00 PM

B9-L2 Room 2223

Presented by **Muhammed Ikbal**

TODAY · 1:00 – 3:00 PM

Two hours, two parts

PART 1 · 1:00 – 2:00

CFD & FDM fundamentals

1. **What is CFD?**
   ingredients, equations, why numerical methods
2. **The 7 pillars of CFD**
   the full workflow, with a running example
3. **Problem 1 by hand**
   the finite difference method with a calculator

PART 2 · 2:00 – 3:00

Hands-on COMSOL

1. **Problem 1 · Pin fin**
   1D steady heat transfer
2. **Problem 2 · Phosphorus diffusion**
   1D time-dependent transport
3. **Problem 3 · Square plate**
   2D conduction and a mesh study

… and more problems from the repo if time allows

Everything is in our repository

github.com/maryamomer2004/simple_CFD

Scan to open the repo: PDFs, notebooks and COMSOL models

Every problem folder contains

Problem statement — PDF

Step-by-step solution — PDF

FDM notebook — plain Python

Whiteboard guide — solved by hand

COMSOL model — .mph + guide

7 problems, from 1D conduction to the Navier–Stokes equations

1	2	3	4	5	6	7
Pin fin	**Diffusion**	**Plate**	**Brick column**	**Conv.–diff.**	**Cavity flow**	**Natural conv.**
1D · steady	1D · unsteady	2D · Laplace	2D · Robin BC	1D · Péclet	2D · Navier–Stokes	2D · buoyancy

orange = today's hands-on

PART 1 · 1:00 – 2:00 PM

Introduction to CFD & the Finite Difference Method



What CFD is · the 7 pillars · Problem 1 solved by hand

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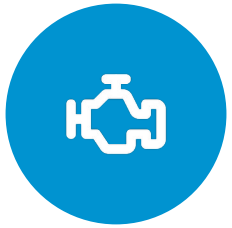
It starts with a physical problem

Something happens in the real world, and we want to **predict** it.



Air over a wing

lift and drag



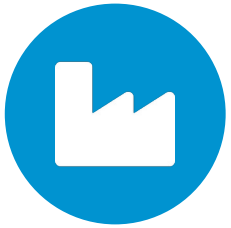
Heat in an engine

hot spots and cooling



Blood in an artery

pressure and wall stress



Smoke from a chimney

where the pollution goes



Computational Fluid Dynamics = predicting flow and heat transfer by solving their equations on a computer

Every problem has three ingredients

Geometry

Where?

A volume of space, the **control volume**, bounded by walls, inlets and outlets.

IN THE EXAMPLE

the inside of a duct with a cylinder in it

+

Physics

What happens?

The event inside that volume: flow, heat transfer, mixing, reactions …

IN THE EXAMPLE

air flows in, speeds up around the cylinder and flows out

+

Math model

How to describe it?

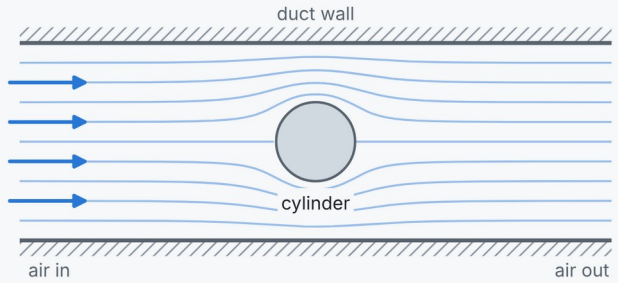
Conservation laws (mass, momentum, energy) as **PDEs**, plus boundary & initial conditions.

IN THE EXAMPLE

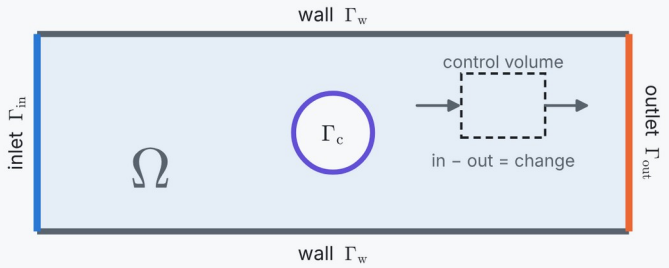
Navier–Stokes, or simply $\nabla^2\psi = 0$

From a physical problem to numbers

1 Physical problem
air flows through a duct, past a cylinder



2 Geometry → domain Ω
the region where the physics happens, and its boundaries



3 Physics → governing equations
conservation laws written as PDEs + boundary conditions

Conservation laws (Navier–Stokes)

mass $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$

momentum $\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$

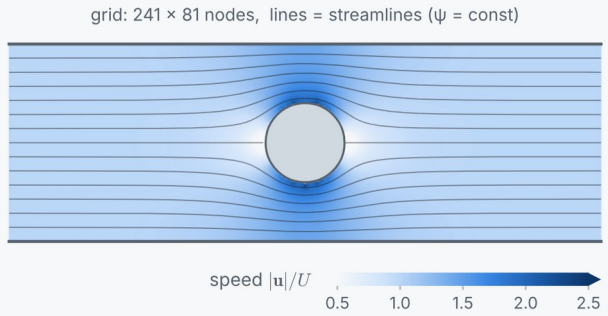
Simplest model: ideal (potential) flow

stream function $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0, \quad u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$

Boundary conditions

$\psi = U y$ on $\Gamma_{in}, \Gamma_{out}$; $\psi = \pm U H$ on Γ_w; $\psi = 0$ on Γ_c

4 Numerical solution
the PDE solved on a grid with finite differences



CFD IN ONE SENTENCE

We take a **geometry** in which a **physical event** happens, describe it with **equations** (PDEs) and, because they can almost never be solved by hand, solve them **approximately, with numerical methods, on a computer.**

Analytical vs numerical solutions

	Analytical	Numerical (CFD)
Geometry	simple: plates, pipes, spheres	arbitrary: cars, aircraft, arteries
Equations	linear, heavily simplified	nonlinear, coupled, 3D, unsteady
Result	a formula, valid everywhere	numbers at a finite set of grid points
Accuracy	exact (for the chosen model)	approximate: error shrinks as the grid is refined
Cost	pen and paper	computer time and memory

Analytical solutions are still precious: they are how we **check** a CFD code.

One equation, many physics

Most CFD equations are special cases of one **generic transport equation** for a quantity φ:

$$\frac{\partial(\rho\phi)}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\,\phi) = \nabla\cdot(\Gamma\,\nabla\phi) + S_\phi$$

rate of change	convection: carried by the flow	diffusion: spreading	source

Choose φ =

φ = 1

mass · continuity

φ = u, v, w

momentum · Navier–Stokes

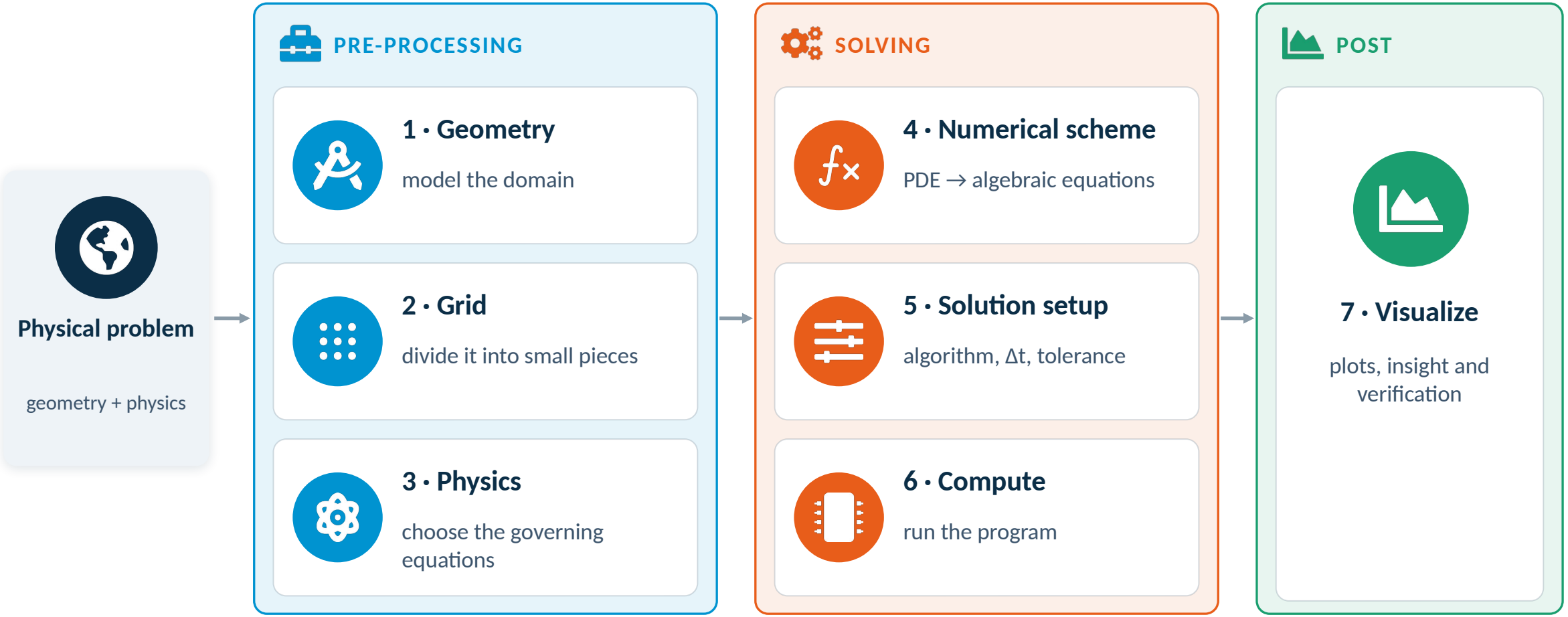
φ = T

energy · heat transfer

φ = c

a species · mixing, pollution

Every CFD analysis rests on 7 pillars



Heating a rod: our running example

At t = 0 the whole rod is at 20 °C. Then its left end is suddenly heated to 100 °C, while the right end stays at 20 °C.



How does the temperature inside the rod change with time?

We will follow this rod through all 7 pillars:

1 Geometry
2 Grid
3 Physics
4 Scheme
5 Setup
6 Compute
7 Visualize

Geometry: model the domain

Define **where** the physics happens: the domain Ω and its boundaries. The art is smart **simplification**:

Dimension

Can 3D become 2D, or even 1D? Each dimension removed cuts the cost enormously.

Symmetry

Symmetric geometry and physics: model only half, or a quarter.

Defeaturing

Remove bolts, fillets and logos that do not change the result.

Extent

Put inlets and outlets far enough away not to disturb the region of interest.

1D, 2D or 3D?

1D — **Heat conduction along a thin rod**

a line, $0 \le x \le L$

2D — **Flow around a long cylinder, a heated plate**

a plane region

3D — **Flow around a car**

a volume

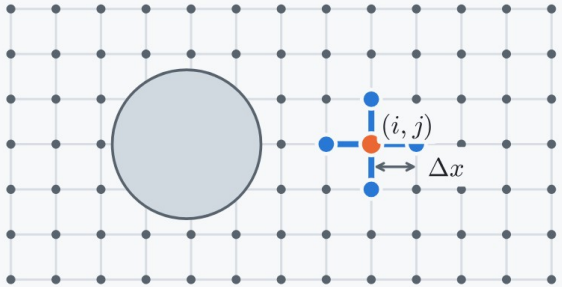
Rod: long and thin, so T varies only along its length → a **1D domain** $0 \le x \le L$, hot end at $x = 0$, cold end at $x = L$.

Grid: divide the domain into pieces

Finite Difference (FDM)

THIS COURSE

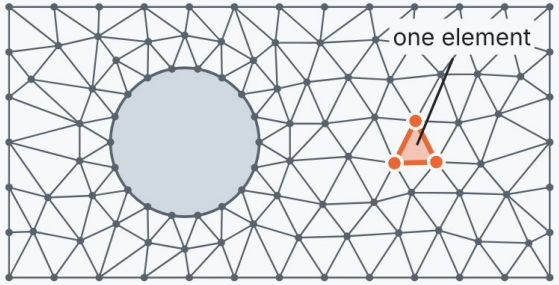
unknowns live on grid points



derivatives → differences of nodal values

Finite Element (FEM)

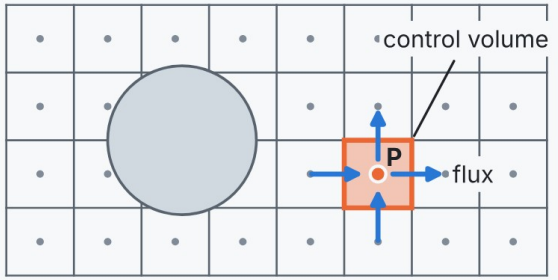
unknowns live on element nodes



simple shape functions on every element

Finite Volume (FVM)

unknowns are cell averages



fluxes balanced across every cell face

Node: where an unknown lives, index i or (i, j)

Grid spacing: Δx, Δy between neighbouring nodes

Structured / unstructured: rows & columns, or any shape

Grid independence: refine until the answer stops changing

Rod: N + 1 equally spaced nodes, $x_i = i\Delta x$ with $\Delta x = L/N$. With N = 20, the whole rod is just **21 temperatures**.

Three families of numerical methods

Finite Difference (FDM)

THIS COURSE · PART 1

UNKNOWNS AT
grid points (nodes)

EQUATIONS IN
differential form

KEY IDEA
derivatives → differences (Taylor)

STRENGTHS
simple, intuitive, easy to code

TYPICAL USE
teaching and research codes

Finite Element (FEM)

COMSOL · PART 2

UNKNOWNS AT
element nodes

EQUATIONS IN
weak (weighted-integral) form

KEY IDEA
simple functions on every element

STRENGTHS
complex geometries, rigorous theory

SOFTWARE
COMSOL, Abaqus, FEniCS

Finite Volume (FVM)

FLUENT · OPENFOAM

UNKNOWNS AT
cell centres (cell averages)

EQUATIONS IN
integral (conservation) form

KEY IDEA
balance fluxes through cell faces

STRENGTHS
conserves mass, momentum, energy

SOFTWARE
Fluent, OpenFOAM, STAR-CCM+



We **learn** with FDM: the most direct translation of calculus into code. Grids, discretization, stability and convergence carry straight over to FEM and FVM.

Physics: choose the governing equations

Ask yourself

- Steady or unsteady?
- Compressible or incompressible?
- Viscous? Laminar or turbulent?
- Does heat transfer matter?
- Buoyancy, reactions, phases?

Every "yes" adds terms or equations; every justified "no" makes the problem simpler.

MODEL EQUATION	FORM	TYPE
Laplace steady conduction, ideal flow	$\nabla^2 \phi = 0$	elliptic
Poisson heat sources, pressure equation	$\nabla^2 \phi = f$	elliptic
Heat / diffusion unsteady conduction, diffusion	$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$	parabolic
Linear advection a quantity carried by a flow	$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$	hyperbolic
Burgers a 1D "toy" Navier–Stokes	$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$	mixed
Navier–Stokes general viscous fluid flow	$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{u}$	mixed

Elliptic: every point influences every other · **Parabolic:** marches in time while smoothing · **Hyperbolic:** carried along directions at finite speed

Boundary & initial conditions

A PDE alone has infinitely many solutions. Boundary conditions (and, if unsteady, initial conditions) pick the one that matches your problem.

Dirichlet

prescribes the value

$$T = T_w$$

e.g. a wall held at a fixed temperature, a known inlet velocity

Neumann

prescribes the gradient

$$\frac{\partial T}{\partial n} = g$$

e.g. an insulated wall (∂T/∂n = 0), a symmetry plane

Robin (mixed)

a combination of both

$$-k\,\frac{\partial T}{\partial n} = h\,(T - T_\infty)$$

e.g. a surface cooled by the surrounding air

Rod: the 1D heat equation

α = k/(ρc$_p$): thermal diffusivity

$$\frac{\partial T}{\partial t} = \alpha\,\frac{\partial^2 T}{\partial x^2}$$

$$T(0,t) = 100\,^\circ\mathrm{C} \quad T(L,t) = 20\,^\circ\mathrm{C} \quad T(x,0) = 20\,^\circ\mathrm{C}$$

Numerical scheme: derivatives → differences

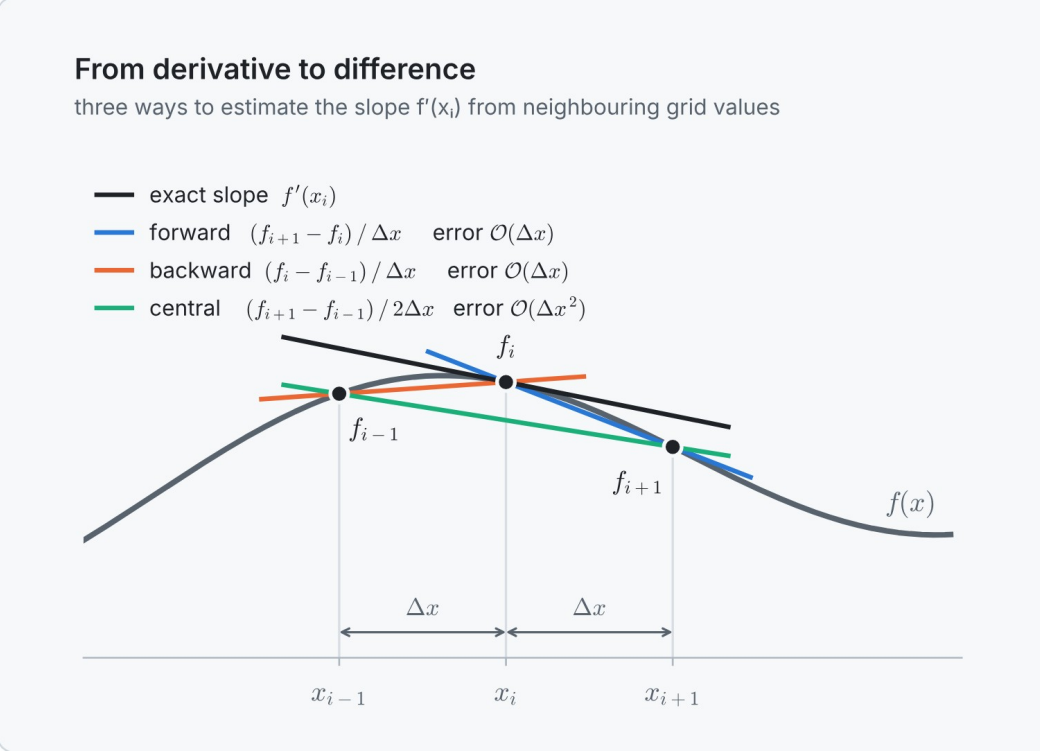
1 · TAYLOR SERIES AROUND x_i

$$f(x_i + \Delta x) = f_i + \Delta x\, f'_i + \frac{\Delta x^2}{2} f''_i + \frac{\Delta x^3}{6} f'''_i + \cdots$$

2 · SOLVE FOR f'_i: THE FORWARD DIFFERENCE

$$f'_i = \frac{f_{i+1} - f_i}{\Delta x} \underbrace{- \frac{\Delta x}{2} f''_i - \cdots}_{\text{truncation error, O(Δx)}}$$

The terms we throw away are the **truncation error**. The power of Δx in its leading term is the **order of accuracy** p.



Three ways to estimate the slope from neighbouring grid values. The central difference is closest to the exact tangent.

The finite difference toolbox

f'_i · forward — 1st order

$$\frac{f_{i+1} - f_i}{\Delta x}$$

f'_i · backward — 1st order

$$\frac{f_i - f_{i-1}}{\Delta x}$$

f'_i · central — 2nd order

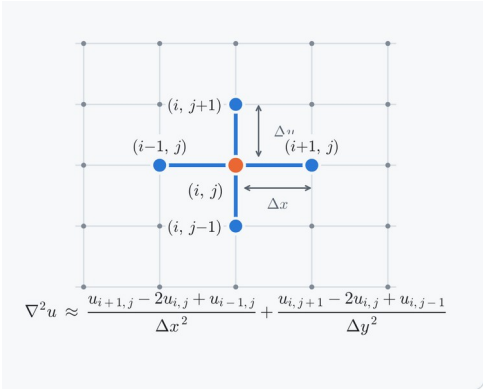
$$\frac{f_{i+1} - f_{i-1}}{2\,\Delta x}$$

f''_i · central — 2nd order

$$\frac{f_{i+1} - 2f_i + f_{i-1}}{\Delta x^2}$$

÷4

Halve Δx and a **2nd-order** error falls 4×; a 1st-order error only 2×. In general, error ∝ Δx^p.



In 2D, the **five-point stencil** links node (i, j) to its four neighbours.

→ Write one formula at every node → the PDE becomes a set of **algebraic equations**, each linking a node to its neighbours.

The rod becomes plain arithmetic

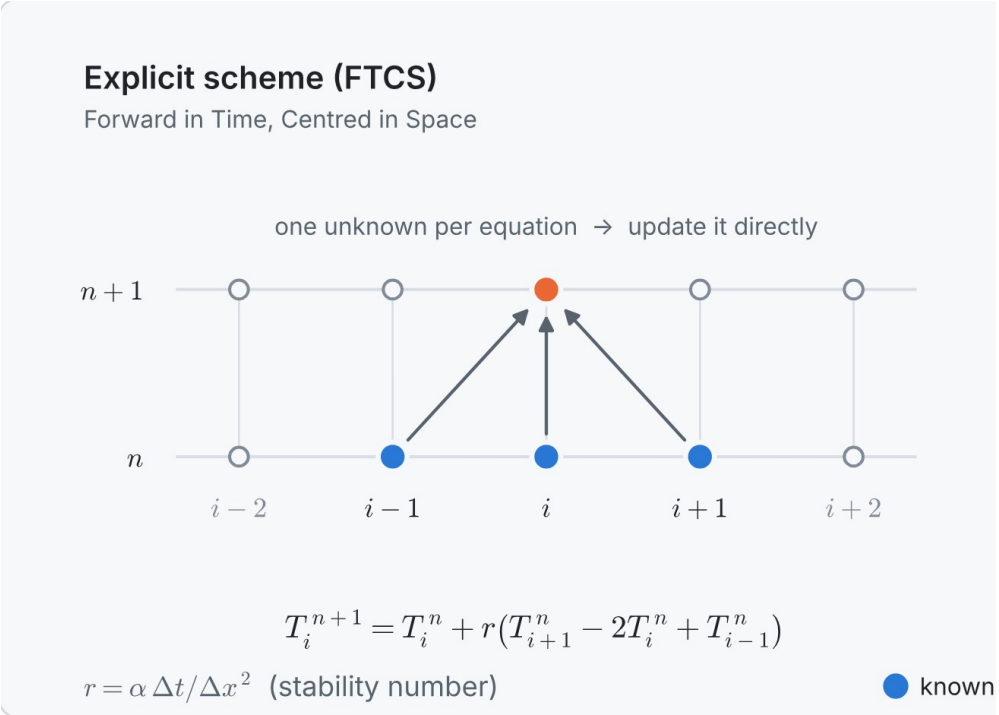
FORWARD IN TIME + CENTRED IN SPACE = FTCS

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}$$

$$T_i^{n+1} = T_i^n + r\left(T_{i+1}^n - 2T_i^n + T_{i-1}^n\right)$$

$$r = \frac{\alpha \Delta t}{\Delta x^2}$$ the stability number (Pillar 5)

No calculus left: each new temperature = old value + r × (sum of neighbour differences).

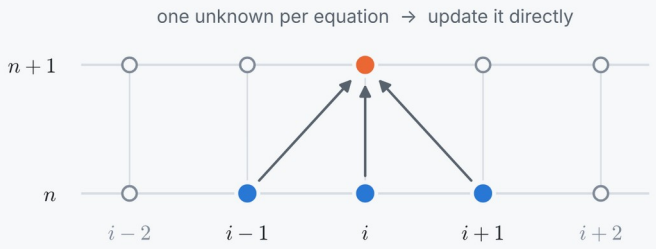


3 known values at time level n → 1 new value at n + 1: every node is updated directly.

Solution setup ①: explicit or implicit?

Explicit scheme (FTCS)

Forward in Time, Centred in Space



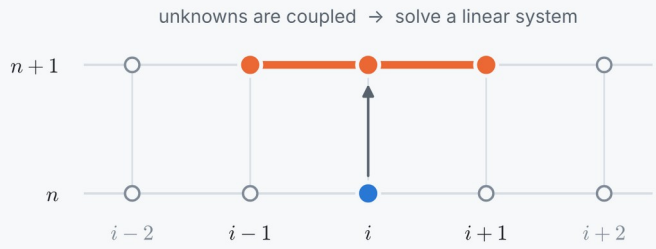
$$T_i^{n+1} = T_i^n + r(T_{i+1}^n - 2T_i^n + T_{i-1}^n)$$

$r = \alpha \Delta t / \Delta x^2$ (stability number)

known

Implicit scheme (BTCS)

Backward in Time, Centred in Space



$$-r\,T_{i-1}^{n+1} + (1+2r)\,T_i^{n+1} - r\,T_{i+1}^{n+1} = T_i^n$$

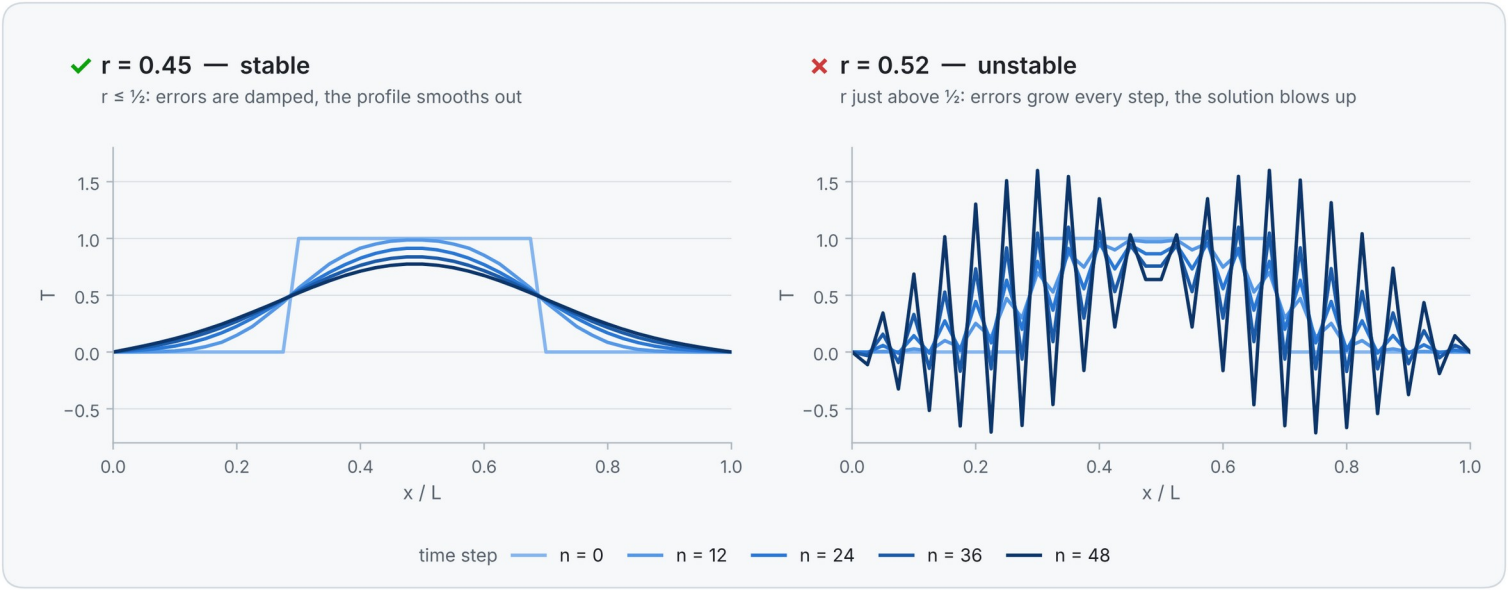
value (time level n)

unknown value (time level n+1)

- new values from old values only
- tiny work per step: just arithmetic
- a few lines of code
- Δt limited by stability (r ≤ ½)

- new values from old values and each other
- solve a linear system every step
- needs a matrix solver
- any Δt is stable: choose it for accuracy

Solution setup ②: is the scheme stable?



FTCS · DIFFUSION

$$r = \frac{\alpha\,\Delta t}{\Delta x^2} \leq \frac{1}{2}$$

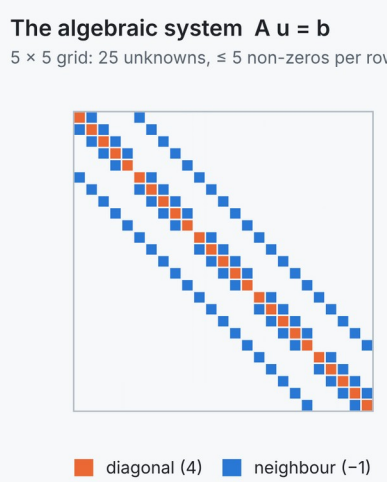
EXPLICIT CONVECTION · CFL

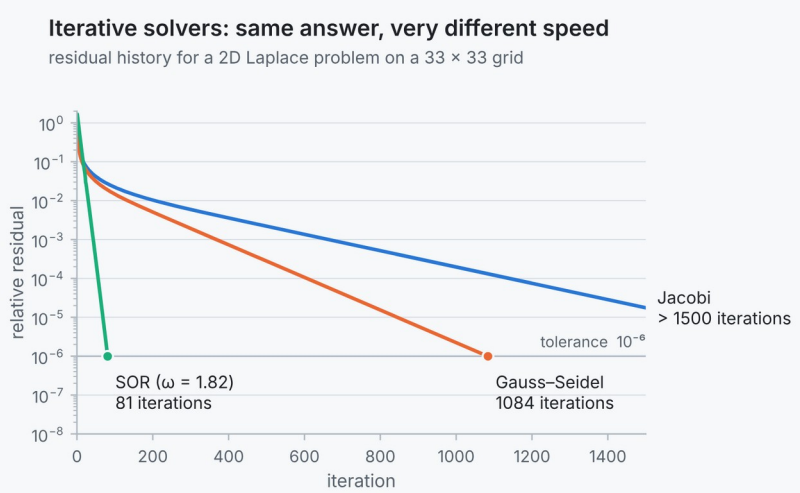
$$\frac{|c|\,\Delta t}{\Delta x} \leq 1$$

Consistency + **Stability** = **Convergence**

Lax equivalence theorem: a scheme approaches the true solution as Δx, Δt → 0 only if it is consistent **and** stable. Exceed the stability limit by a few percent and the solution explodes (right plot).

Solution setup ③: direct or iterative?





$$A\,\mathbf{u} = \mathbf{b}$$

Implicit schemes and steady problems give a linear system. **A** is huge but **sparse**: each equation involves only a node and its neighbours.

Direct

Gauss, LU, **Thomas**: exact in one pass; Thomas costs only O(N)

Jacobi

neighbours' **old** values: simple, but very slow

Gauss–Seidel

uses **new** values at once: ~2× faster than Jacobi

SOR

over-corrects with 1 < ω < 2: often **10–100×** faster

Rod: explicit FTCS with r = 0.4 ≤ ½: simple and stable. (Implicit BTCS would need the Thomas algorithm every step, but allows a larger Δt.)

Compute: run the program

Python
learning, prototyping: this course

Excel / Calc
one cell = one node

MATLAB / Octave
matrix-friendly teaching & research

C++ / Fortran / Julia
large, fast production codes

OpenFOAM · SU2
open-source CFD packages

Part 2

Fluent · COMSOL
commercial, graphical workflow

Same algorithm, many environments.

ROD SOLVER: ABOUT A DOZEN LINES OF PYTHON

```
import numpy as np
import matplotlib.pyplot as plt

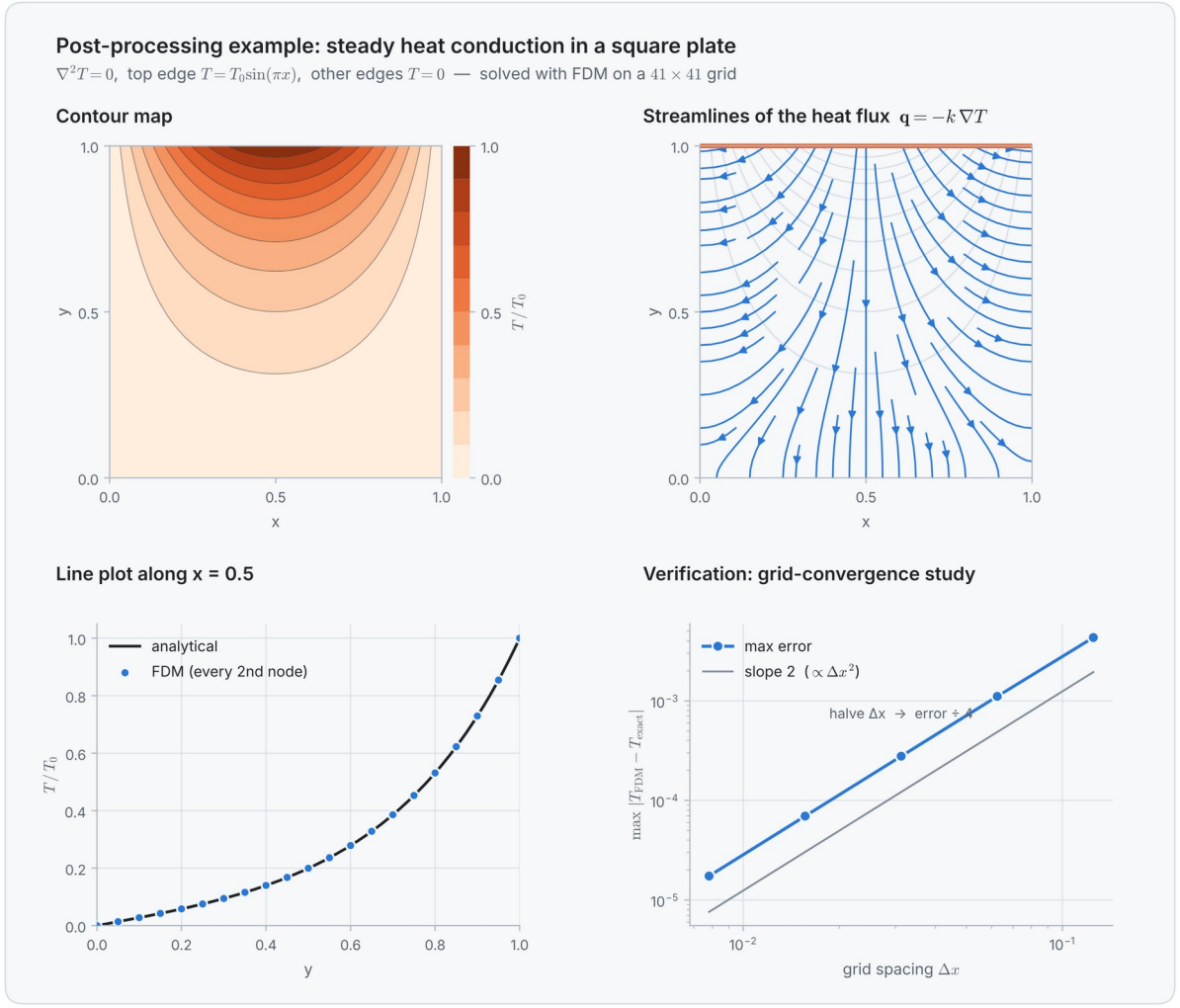
L, N, alpha = 1.0, 20, 1e-4   # rod, intervals, α
dx = L / N                    # pillar 2: grid
r  = 0.4                      # pillar 5: r ≤ 0.5
dt = r * dx**2 / alpha

x = np.linspace(0, L, N + 1)  # 21 nodes
T = np.full(N + 1, 20.0)      # pillar 3: 20 °C
T[0] = 100.0                  # hot left end

for n in range(600):          # pillars 4 + 6
    T[1:-1] += r*(T[2:] - 2*T[1:-1] + T[:-2])

plt.plot(x, T, "o-")          # pillar 7
```

Visualize: turn numbers into insight



Contour / colour map

a scalar field: hot spots, pressure regions

Vectors / streamlines

flow patterns, recirculation, heat paths

Line plot (profile)

compare with theory, experiments, other grids

Animation

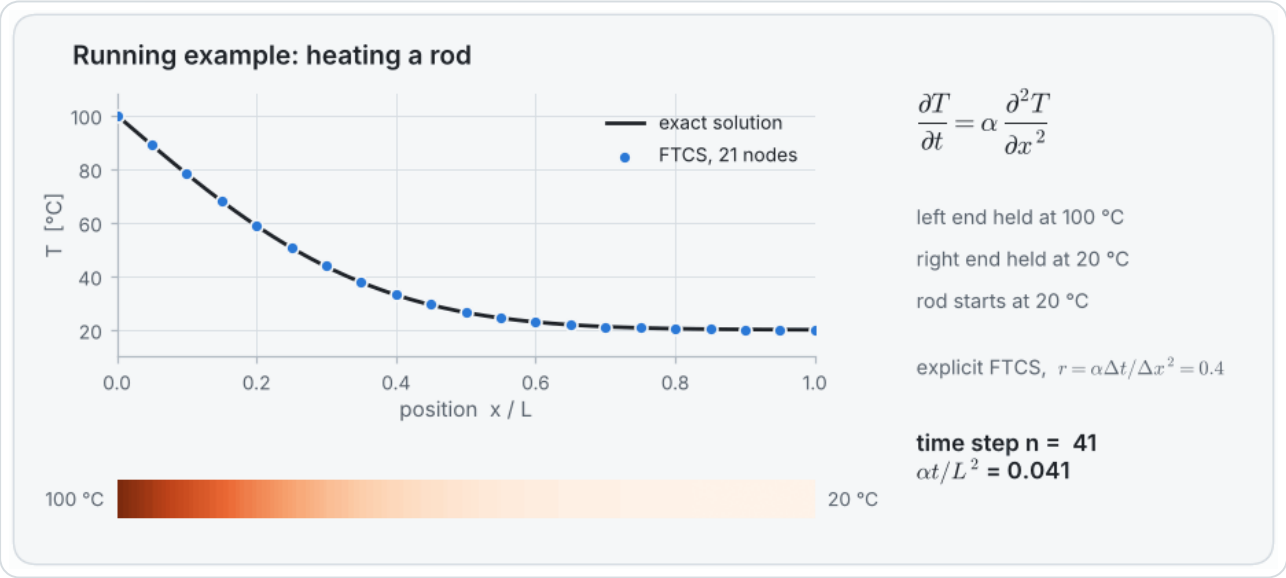
how an unsteady solution evolves

Convergence plot

show that the solution can be trusted

Tools: **Matplotlib** (this course) · ParaView · VisIt · Tecplot · built-in post-processors

Never trust a colourful picture blindly



Rod: 21-node FTCS (dots) on top of the exact solution (line). The rod settles to a straight line from 100 °C to 20 °C.

Verify with: exact solutions | grid refinement | benchmarks

Verification

Are we solving the equations right?

Compare with exact solutions; refine the grid and check the order of accuracy.

Validation

Are we solving the right equations?

Compare with experiments and measurements.

WORKED EXAMPLE • ABOUT 20 MINUTES

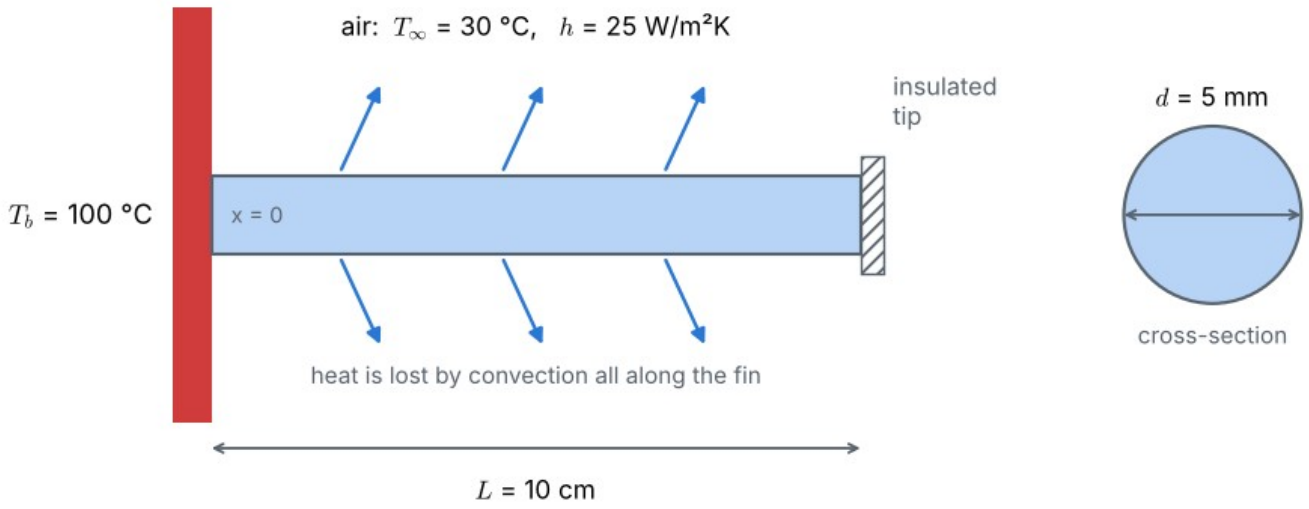
Problem 1: the aluminium pin fin, solved by hand



4 intervals, 4 unknowns and a pocket calculator: every step of the finite difference method

AIChE
KAUST Student Chapter

A pin fin cooled by air



FIND

- the steady temperature T(x)
- the tip temperature
- the heat lost by the fin

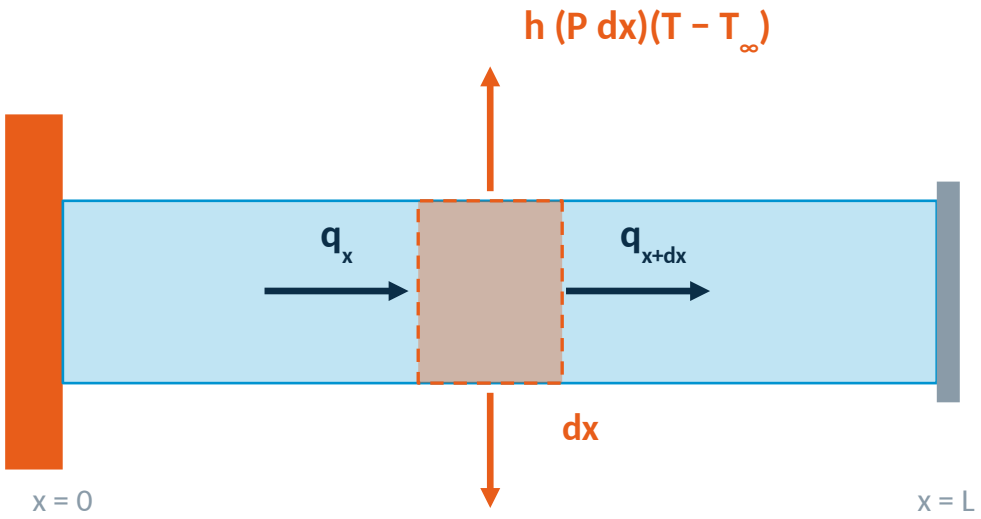
ASK THE CLASS

Where is the fin hottest? Where is it coldest?

Why does the tip not reach 30 °C?

L = 10 cm	d = 5 mm	k = 237 W/(m·K)	h = 25 W/(m²·K)
T_∞ = 30 °C	T_b = 100 °C	insulated tip	steady state

Energy balance → the fin equation



Steady state: **heat in = heat out + heat convected to the air**

θ(0) = 100 − 30 = 70 K (base)

dθ/dx = 0 at x = L (insulated tip)

ENERGY BALANCE ON THE SLICE

$$q_x = q_{x+dx} + h\,(P dx)\,(T - T_\infty)$$

FOURIER'S LAW

$$q_x = -k A_c \frac{dT}{dx}$$

⇒ THE FIN EQUATION

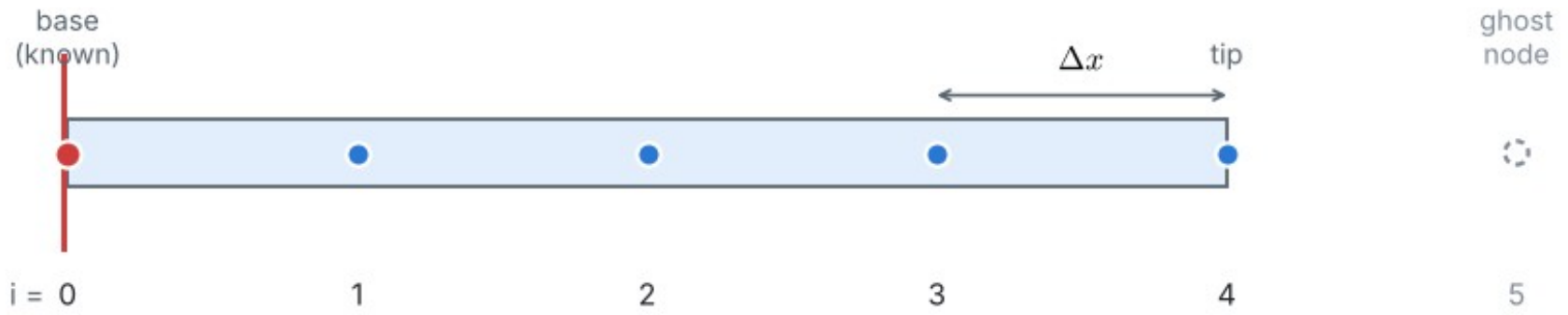
$$\frac{d^2T}{dx^2} - m^2\,(T - T_\infty) = 0 \quad m^2 = \frac{hP}{kA_c} = \frac{4h}{kd}$$

TRICK: EXCESS TEMPERATURE θ

$$\theta = T - T_\infty \quad \Rightarrow \quad \theta'' - m^2\,\theta = 0$$

$m^2 = 4 \times 25 / (237 \times 0.005) =$ **84.388 m^{-2}**

4 intervals, 5 nodes and a ghost node



Node 5 is a **ghost node** beyond the tip: a helper we use to apply the insulated-tip condition.

N = 4
intervals

Δx = 0.025 m
= L / N

θ_0 = 70 K
base node: known

θ_1 … θ_4
the 4 unknowns

Central difference → one equation per node

INTERIOR NODES i = 1, 2, 3

$$\frac{\theta_{i-1} - 2\theta_i + \theta_{i+1}}{\Delta x^2} - m^2 \theta_i = 0$$

↓

$$\theta_{i-1} - B\,\theta_i + \theta_{i+1} = 0$$

$$B = 2 + m^2 \Delta x^2 \qquad = 2 + 84.388 \times 0.025^2 = \mathbf{2.052743}$$

TIP NODE i = 4 (INSULATED)

The tip equation needs θ_5: use the ghost node.

$$\frac{\theta_5 - \theta_3}{2\,\Delta x} = 0 \quad \Rightarrow \quad \theta_5 = \theta_3$$

↓

$$2\,\theta_3 - B\,\theta_4 = 0$$

θ_5 replaces θ_3, so θ_3 appears **twice**: the factor 2.

Calculator tip: keep at least 4 decimals. Rounding B = 2.052743 to 2.05 already shifts the tip temperature by almost 1 °C.

Four equations, one tridiagonal system

$$\begin{bmatrix} B & -1 & 0 & 0 \\ -1 & B & -1 & 0 \\ 0 & -1 & B & -1 \\ 0 & 0 & -2 & B \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{bmatrix} = \begin{bmatrix} 70 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

with B = 2.052743

Each equation links a node only to its **neighbours**: only 3 diagonals are non-zero, a **tridiagonal** system.

The **−2** comes from the ghost node $\theta_5 = \theta_3$.

The known base value $\theta_0 = 70$ moves to the **right-hand side**.

Ask the class: why does each equation contain only neighbouring nodes?

Elimination by hand: the Thomas algorithm

IDEA $\theta_i = P_i\,\theta_{i+1} + Q_i$ write every unknown in terms of the next one: sweep base → tip, then come back tip → base

① Forward sweep: base → tip

i	P_i	Q_i
1	0.487153	34.1007
2	0.638737	21.7814
3	0.707211	15.4040

TIP: PUT θ_3 INTO EQ. (4)

$$\theta_4 = \frac{2\,Q_3}{B - 2P_3} = \frac{30.8081}{0.638321} = 48.264$$

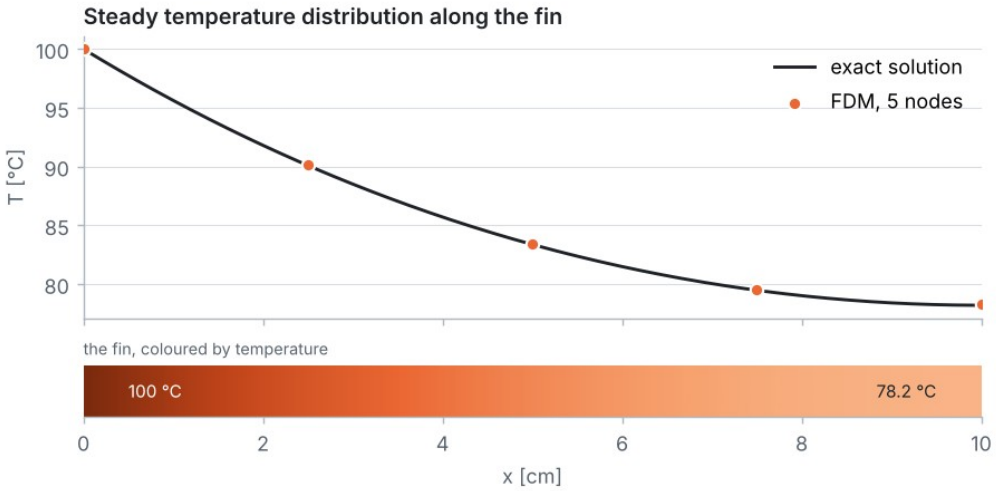
② Back substitution: tip → base

node	θ_i [K]	T_i [°C]
4	48.264	**78.26**
3	49.537	**79.54**
2	53.422	**83.42**
1	60.126	**90.13**

Check one equation, e.g. (2): −60.126 + 2.052743 × 53.422 − 49.537 ≈ 0

By hand, within 0.07 °C of the exact answer

node	x [cm]	by hand [°C]	exact [°C]	error [°C]
0	0	**100.00**	100.00	0
1	2.5	**90.13**	90.09	+0.03
2	5.0	**83.42**	83.37	+0.05
3	7.5	**79.54**	79.47	+0.07
4	10.0	**78.26**	78.19	+0.07



EXACT SOLUTION

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh\, m(L-x)}{\cosh\, mL}$$

m = 9.1863 m^{-1}, mL = 0.9186

0.07 °C
largest error, with only 4 unknowns

2.18 W
heat lost by the fin (exact 2.17 W)

≈ 0.5 %
error of the heat loss

Second-order accuracy, seen by hand

0.28 °C
tip error, 2 intervals

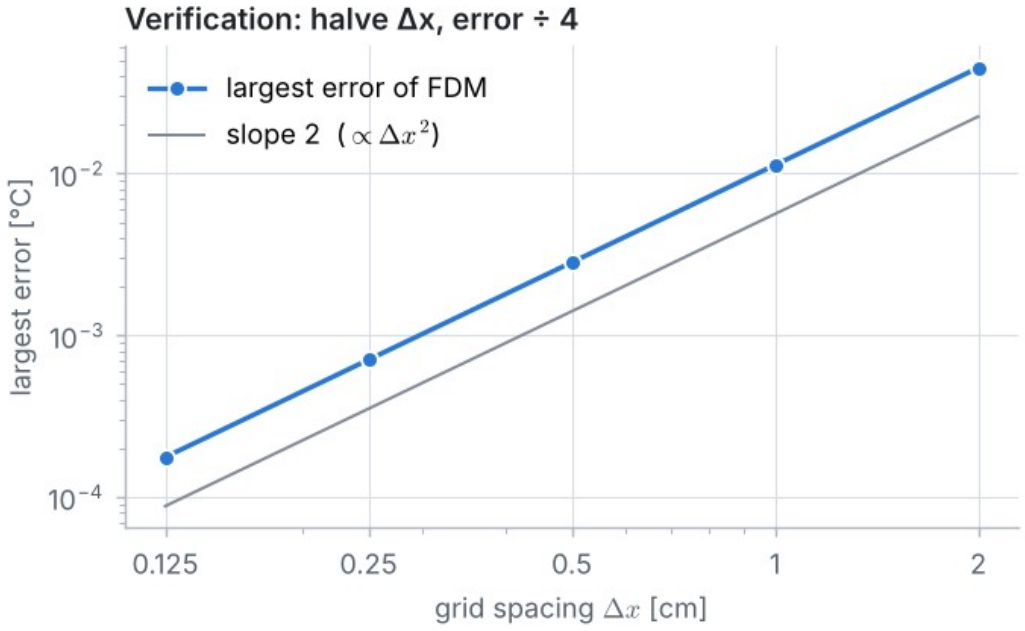
halve Δx →

0.07 °C
tip error, 4 intervals

÷4

Halving Δx divides the error by 4: **second-order accuracy**, exactly as the Taylor series predicted.

2 intervals: B = 2.210970 → T_1 = 83.58 °C, T_{tip} = 78.47 °C (exact 83.37 and 78.19 °C)



The notebook keeps halving Δx: the error falls 4× every time (slope 2 on log axes).

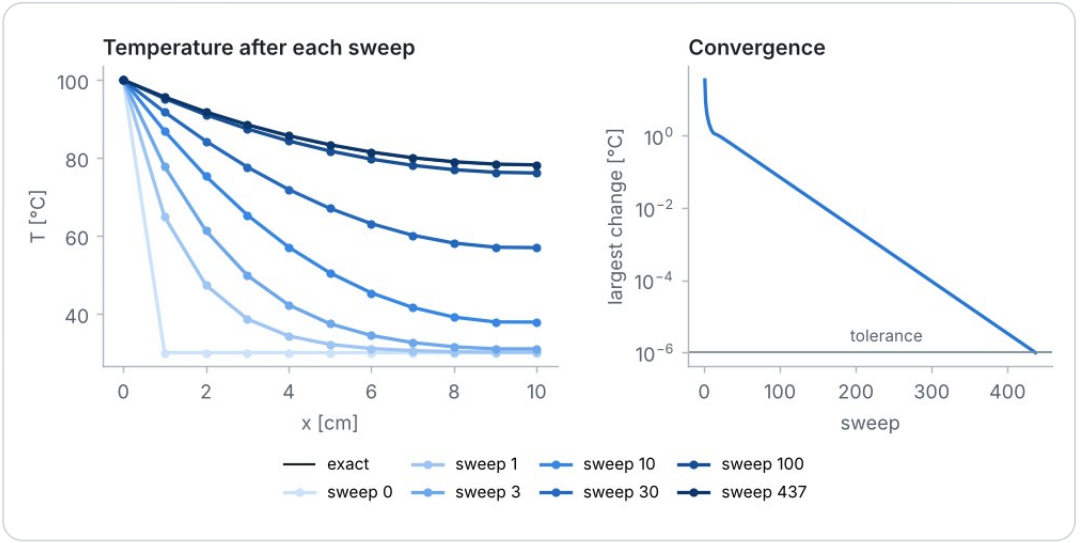
Why we need computers: Gauss–Seidel

SOLVE EACH EQUATION FOR ITS OWN NODE, SWEEP BASE → TIP

$$\theta_1 \leftarrow \frac{70 + \theta_2}{B} \qquad \theta_2 \leftarrow \frac{\theta_1 + \theta_3}{B}$$

$$\theta_3 \leftarrow \frac{\theta_2 + \theta_4}{B} \qquad \theta_4 \leftarrow \frac{2\,\theta_3}{B}$$

sweep	T_1	T_2	T_3	T_4 [°C]
0	30.00	30.00	30.00	30.00
1	64.10	46.61	38.09	37.88
2	72.19	54.50	45.77	45.37
3	76.03	60.11	52.16	51.59
10	86.92	78.08	73.26	72.15
20	89.73	82.77	78.77	77.52
40	90.12	83.41	79.53	78.25
60	**90.13**	**83.42**	**79.54**	**78.26**



≈ 60 sweeps to match the elimination result. An easy rule, painful by hand, and perfect for a computer: milliseconds.

Common mistakes, and from board to notebook

Common mistakes

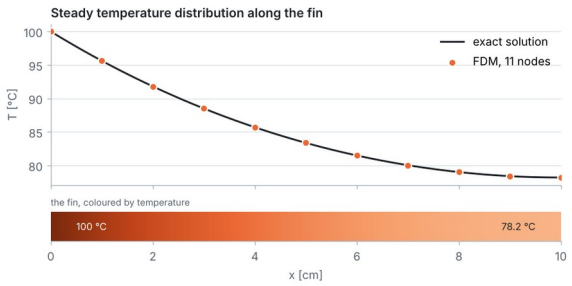
- Forgetting the **factor 2** in the tip equation
- Using Δx in **centimetres** in $m^2\Delta x^2$
- Forgetting T_∞: work with $\theta = T - T_\infty$
- Rounding **B** too early
- Writing an equation for the **base node** (it is known!)

From board to notebook

`03_fdm_solver.ipynb`

Board 4	→	`build_system`
Board 5	→	`thomas`
Board 7	→	`gauss_seidel`

Set **N = 10** (11 nodes):



78.205 °C
tip (exact 78.194)

2.1721 W
heat loss (exact 2.1702)

What we have learned



CFD = PDEs solved on a grid

Geometry, physics and a math model, solved approximately with numerical methods, at a finite set of points.

FDM: derivatives → differences

The Taylor series gives the formulas and their order: halve Δx and a 2nd-order error falls by 4.

Trust needs checks

Stability limits, grid refinement, exact solutions and benchmarks, before any colourful plot.

Next: the same fin in COMSOL, with finite elements →

PART 2 • 2:00 – 3:00 PM

Hands-on COMSOL Multiphysics



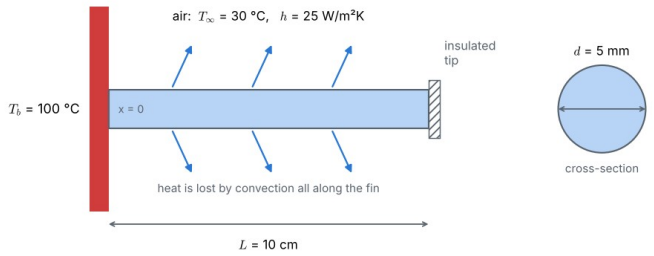
Build, solve and verify the repo's problems step by step, starting with Problem 1

AIChE
KAUST Student Chapter

The 7 pillars inside COMSOL

Model Wizard	space dimension · physics · study type	1 · Geometry · 3 · Physics · 5 · Setup
Global Definitions	parameters: all problem data in one place	problem data
Geometry	interval, square …: the domain	1 · Geometry
Materials	k, ρ, c_p	3 · Physics
Physics nodes	the equation + boundary conditions	3 · Physics · 4 · Scheme (FEM, auto)
Mesh	the elements: our grid	2 · Grid
Study › Compute	stationary or time-dependent solver	5 · Setup · 6 · Compute
Results	plots, point evaluations, integrals	7 · Visualize

From Problem 1, as far as time allows

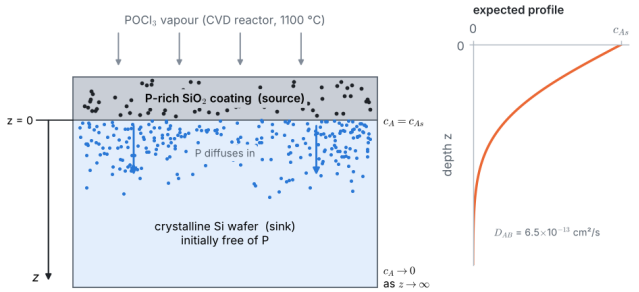


Problem 1 · Pin fin

1D · steady · Heat Transfer in Solids

New: 1D model, heat source, point evaluation

~25 min

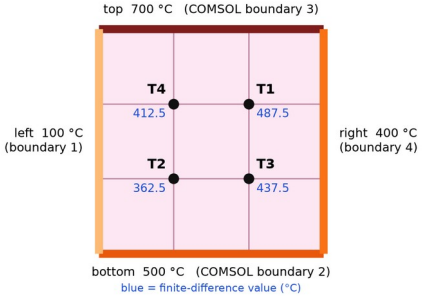


Problem 2 · Diffusion

1D · time-dependent · Transport of Diluted Species

New: parameters, transient study, exact-curve overlay

~15 min



Problem 3 · Square plate

2D · steady · Heat Transfer in Solids

New: 2D geometry, parametric mesh sweep

~15 min

IF TIME ALLOWS · TAKE-HOME

- **4 · Brick column (Robin BC)**
- **5 · Convection–diffusion**
- **6 · Lid-driven cavity**

Before you start

Get the files

```
problems/01_aluminium_fin/
  comsol_solution/
```

`COMSOL_PinFin_Guide.pdf`

12 annotated steps with screenshots

`fin1d_pin_fin.mph`

the finished model, to compare

COMSOL 4.3a or newer

Licence: university network or VPN

Menus

Left-click a tree node first, then right-click it.

Select points & edges

Hover until it turns **red** → click → press **+**.

Units in [brackets]

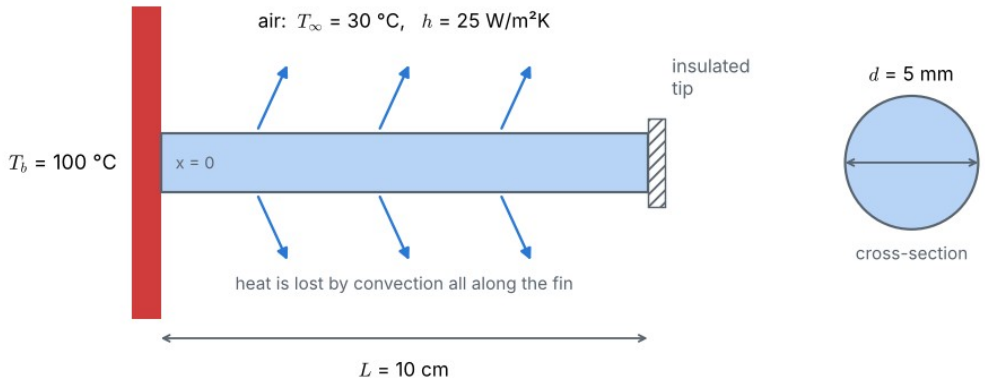
373.15[K]
0.1[um]
6.5e-13[cm^2/s]

Kelvin & metres

100 °C = 373.15 K. Type 0.1 (m), not 10 (cm).

Save often: File › Save As, and compare with the supplied .mph if you get stuck.

Translating the fin into a COMSOL model



TARGET: THE EXACT SOLUTION

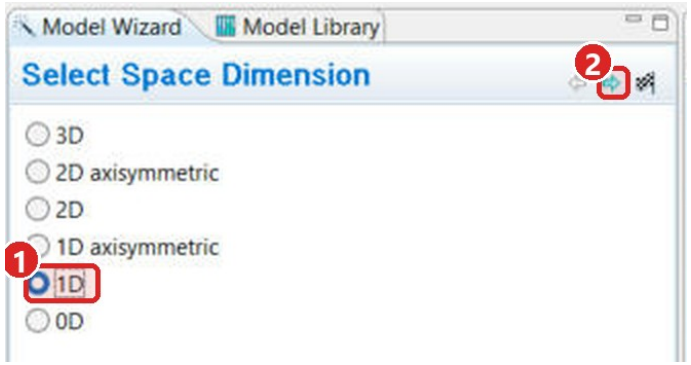
78.194 °C
tip temperature

2.1702 W
heat lost by the fin

PROBLEM	COMSOL FEATURE
The fin, 10 cm long	**Interval** 0 → 0.1 m (1D)
Base at 100 °C	**Temperature** 373.15 K at Point 1
Insulated tip	**Thermal Insulation** at Point 2 (default!)
Convection from the side	**Heat Source** $Q = -h\,(4/d)(T - T_\infty)$
Steady state	**Stationary** study

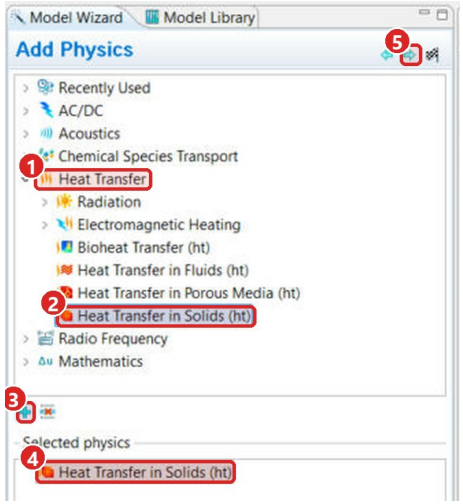
Model Wizard: 1D → physics → study

1 Select **1D**, then Next



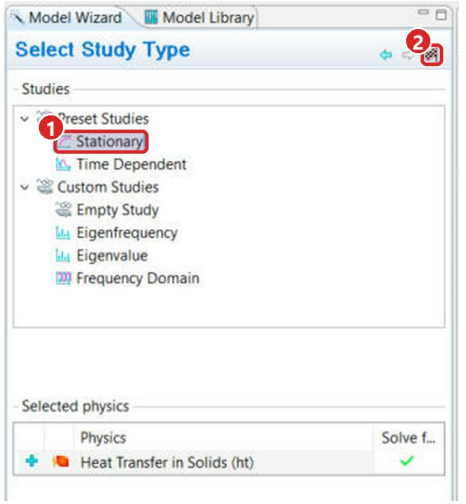
Why? The pin is long and thin: one temperature per cross-section, T(x) only.

2 Heat Transfer › **Heat Transfer in Solids** › **+** › Next



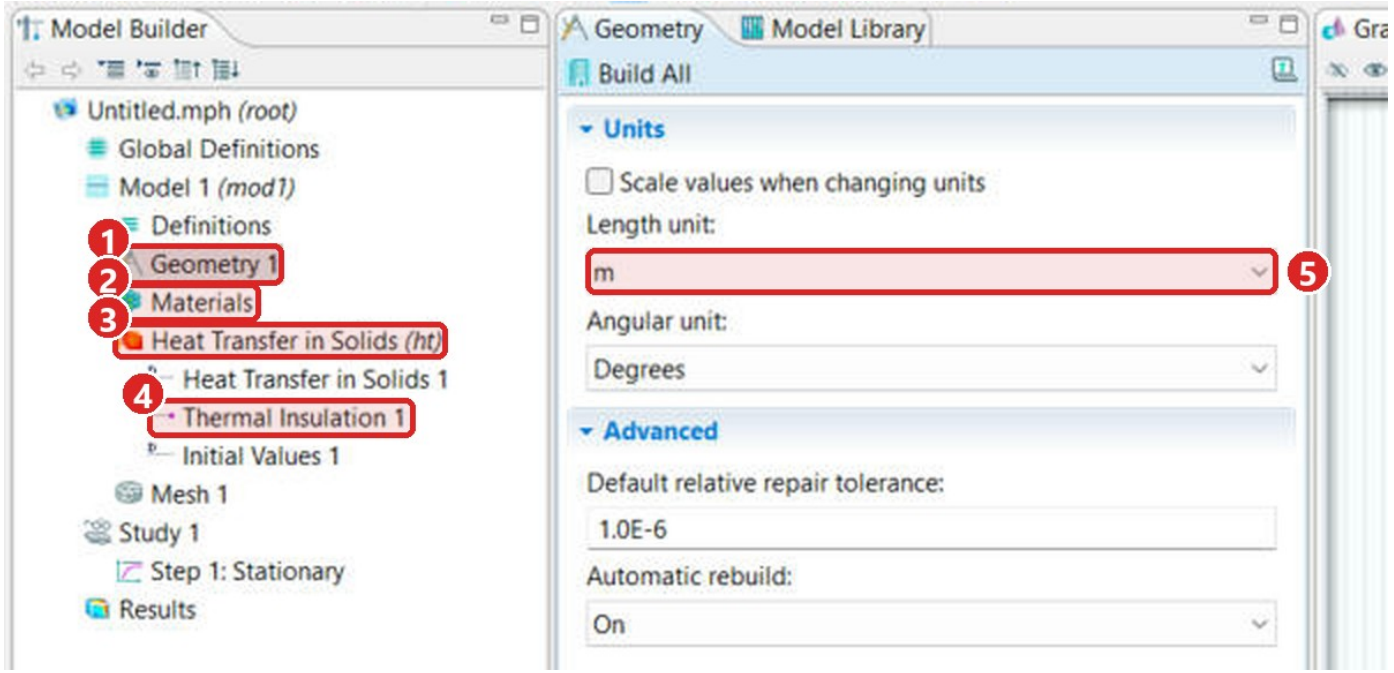
Why? Solves −d/dx(k dT/dx) = Q. We will use Q for the convective loss.

3 **Stationary** › Finish



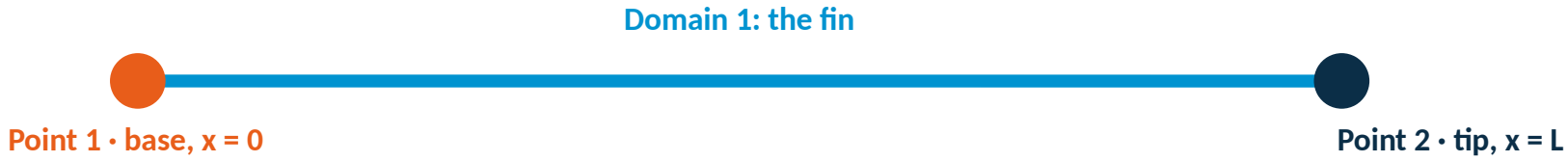
Why? Steady state: the initial 30 °C has no influence, so it is not needed.

A tour of the Model Builder tree



1. **Geometry 1**
   the fin's shape (Step 5)
2. **Materials**
   aluminium (Step 6)
3. **Heat Transfer in Solids**
   equation & BCs (Steps 7–8)
4. **Thermal Insulation 1**
   default on every boundary = our insulated tip
5. **Length unit: m**
   enter all lengths in metres

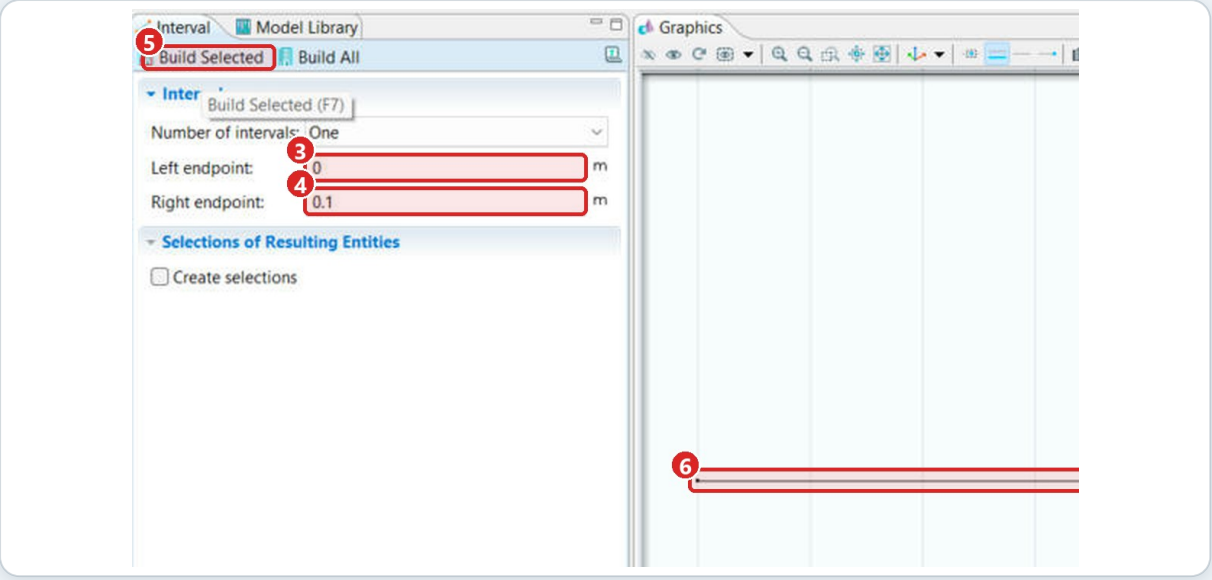
In 1D the domain is a line; its boundaries are its two end points.



Geometry and material

5 Geometry: a 10 cm line

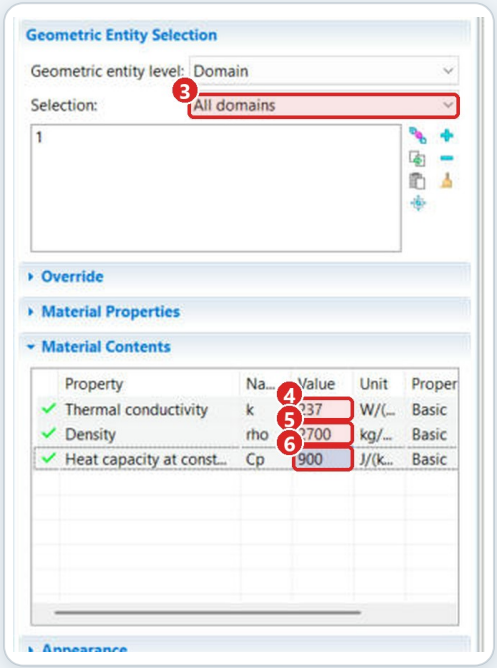
Right-click **Geometry 1** › **Interval** · Left endpoint **0** · Right endpoint **0.1** · **Build Selected** (F8)



Type **0.1**, not 10: the length unit is metres.

6 Material: aluminium

Right-click **Materials** › **Material** (blank) · Selection: **All domains**



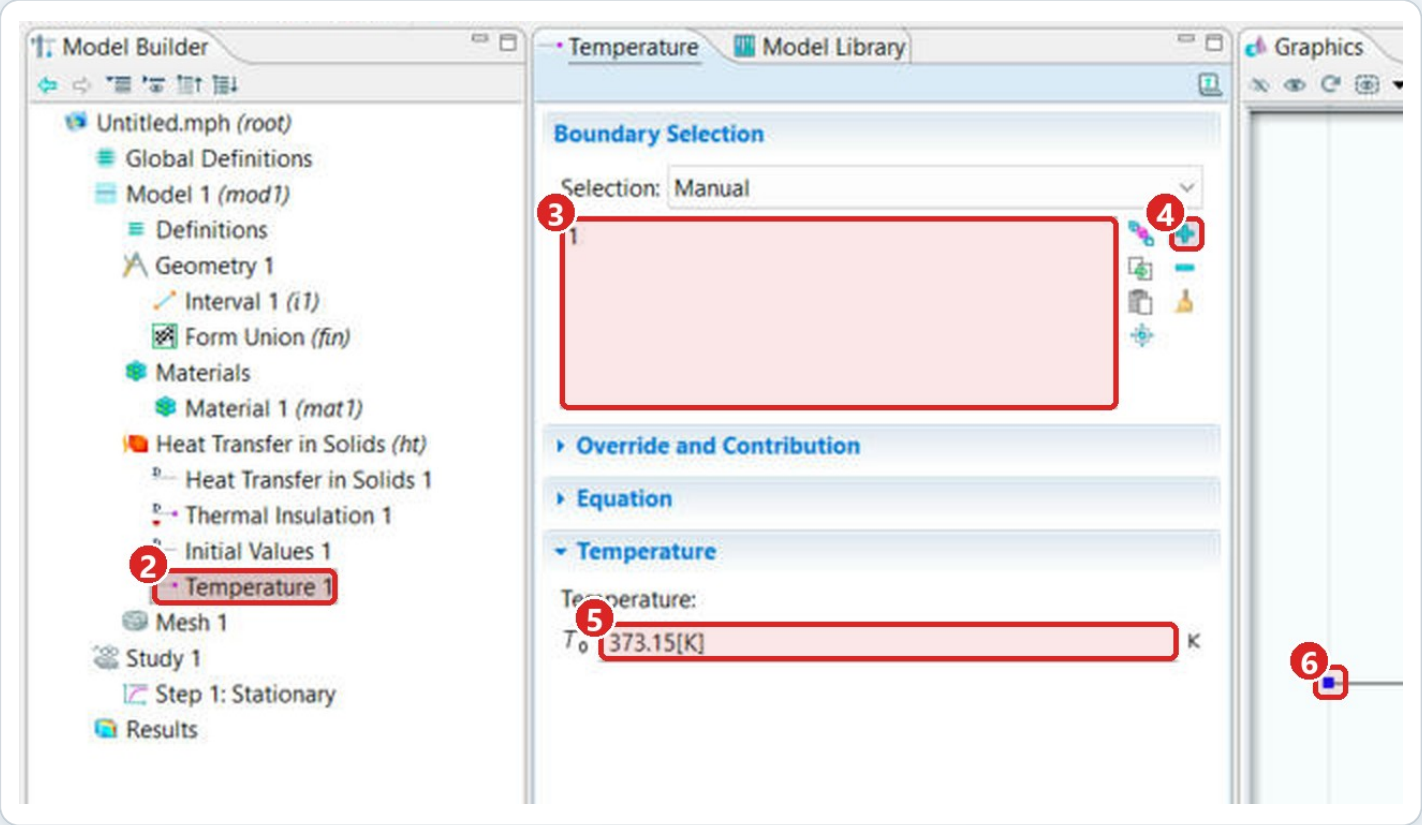
k = 237 W/(m·K)

ρ = 2700 kg/m³

C_p = 900 J/(kg·K)

Only **k** matters for the steady state. COMSOL needs ρ and C_p for a complete material.

Boundary condition: the base at 100 °C

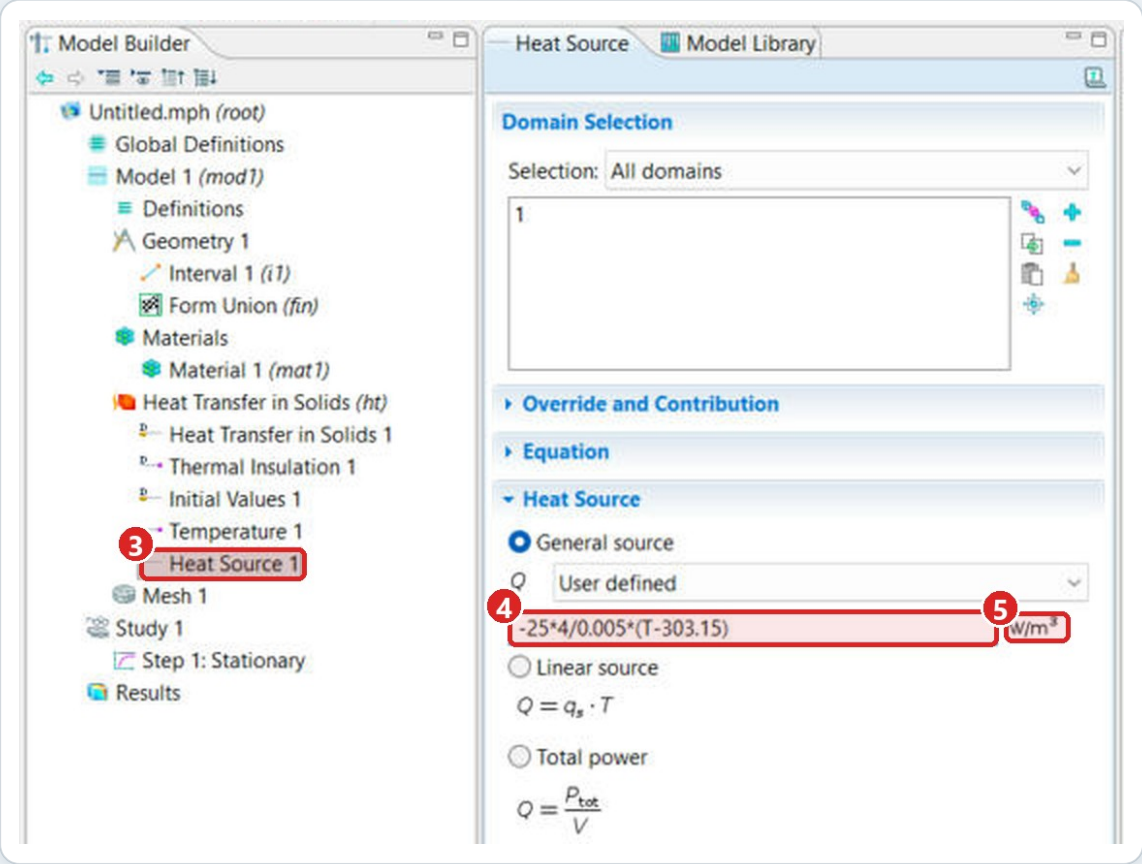


1. Right-click **Heat Transfer in Solids › Temperature**
2. Select **Point 1**: hover until red → click → **+**
3. T_0 = **373.15[K]**

100 °C = 373.15 K

Why? A Dirichlet condition $T(0) = T_b$. Thermal Insulation stays on Point 2: the insulated tip.

Convection as a heat sink



A 1D model has no side surface, so we add the convective loss as a **volumetric heat sink** along the fin.

WHAT COMSOL SOLVES

$$-\frac{d}{dx}\left(k\,\frac{dT}{dx}\right) = Q$$

THE SOURCE: LOSS PER UNIT VOLUME (P/A = 4/d)

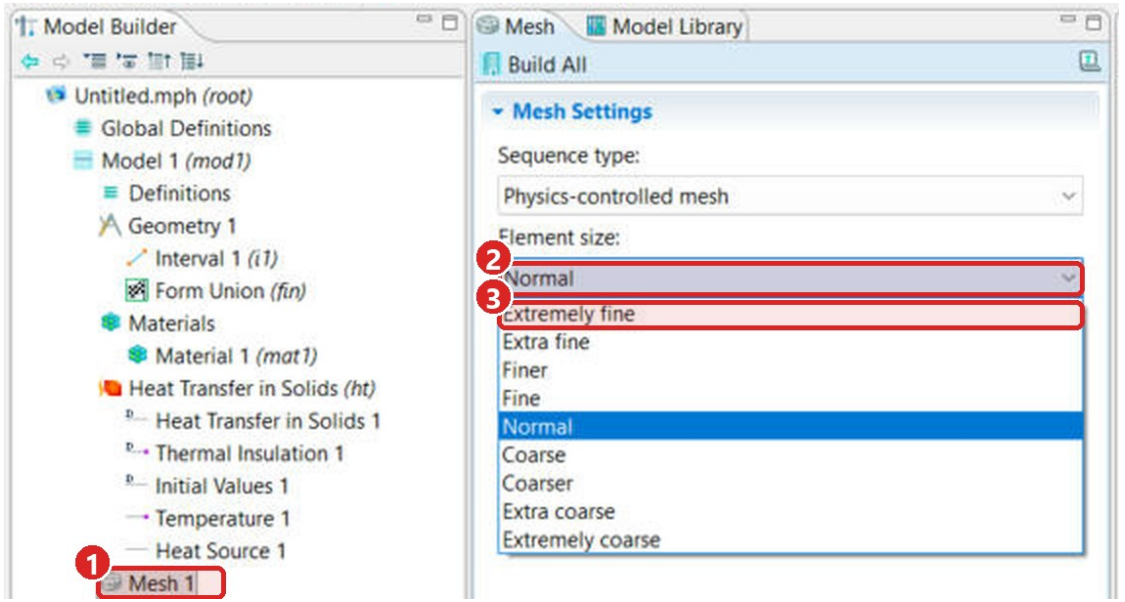
$$Q = -\,h\,\frac{P}{A}\,(T - T_\infty) = -\,h\,\frac{4}{d}\,(T - T_\infty)$$

TYPE IN THE Q FIELD (UNIT W/m³)

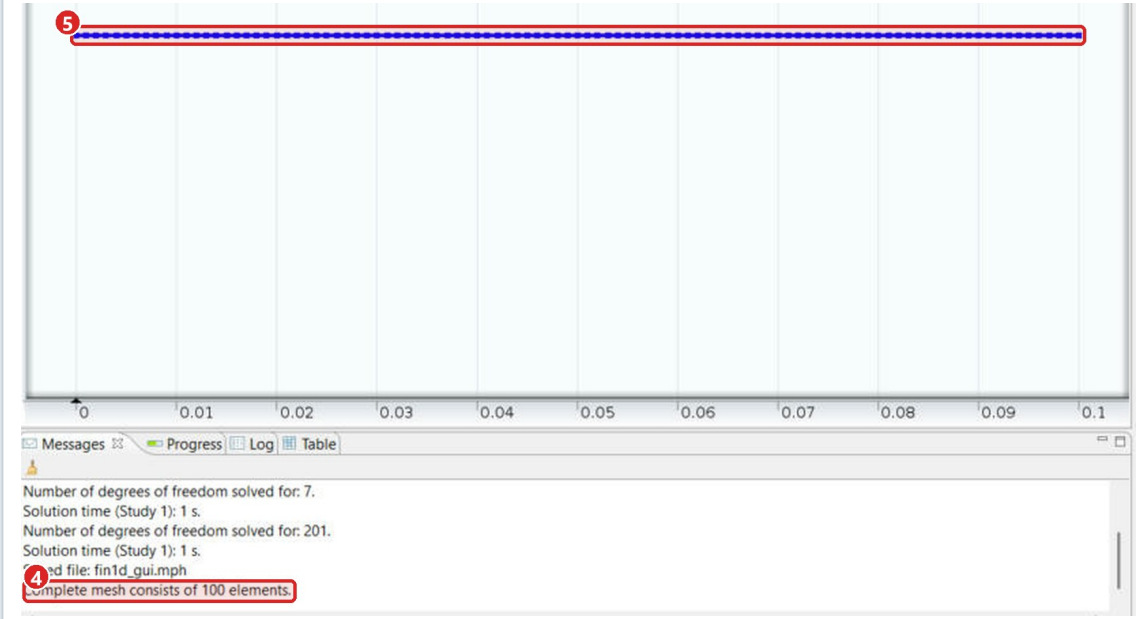
```
-25*4/0.005*(T-303.15)
```

The minus sign makes it a **sink**: heat leaves the fin. 303.15 K = T_∞ = 30 °C.

Mesh: 100 elements along the fin



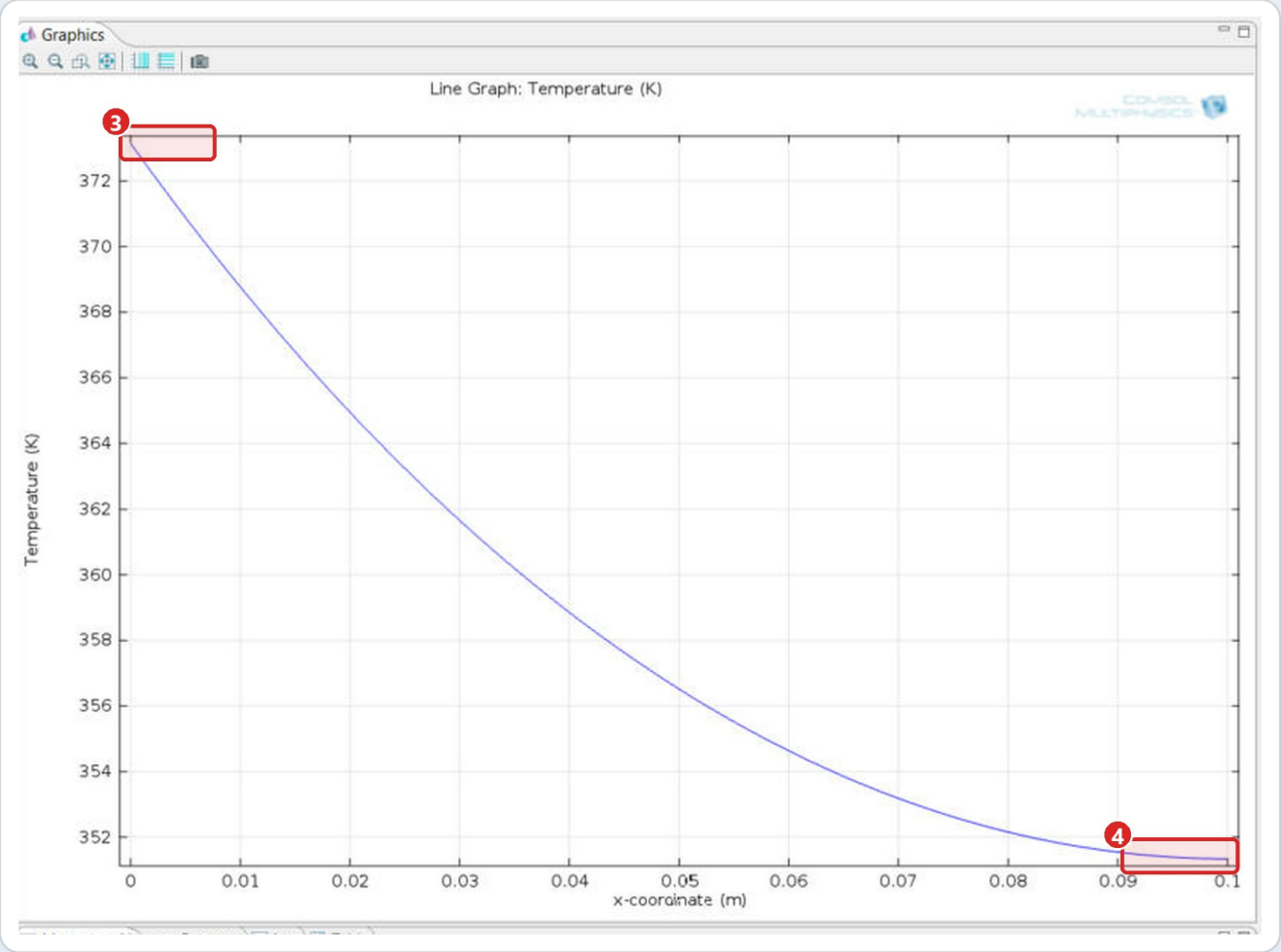
Mesh 1 › Element size: **Extremely fine** › **Build All** (F8)



Messages: **Complete mesh consists of 100 elements** (Δx = 1 mm)

The FEM mesh plays the role of our FDM grid. Later, try **Normal** (15) and **Extremely coarse** (3 elements).

Compute: the temperature profile



10 Right-click **Study 1** › **Compute** (F8)

373.15 K
= 100 °C at the base

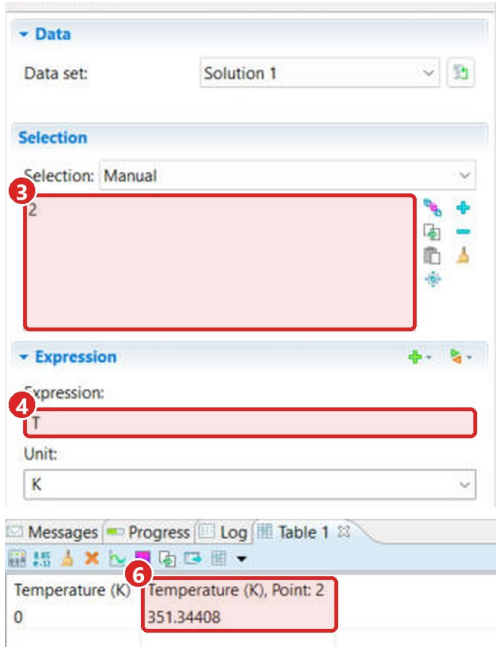
351.3 K
= 78.2 °C at the tip

Steepest slope at the base, where heat enters; **zero slope at the tip** = insulated. The classic fin profile cosh m(L − x) / cosh mL.

Read the numbers: tip temperature and heat loss

11 Tip temperature

Derived Values › **Point Evaluation** · select **Point 2** (the tip) · Evaluate (=)



EXPRESSION

```
T
```

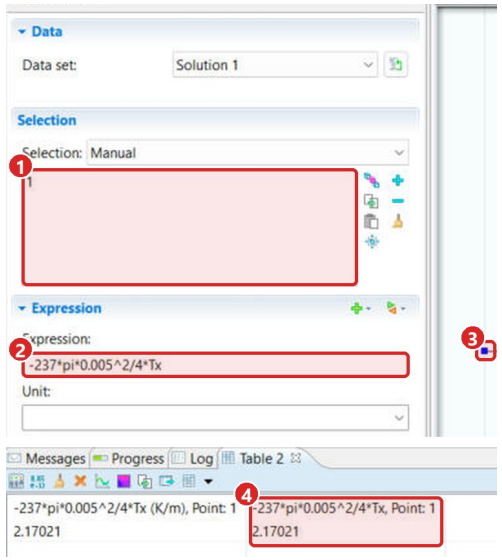
the temperature, in K

RESULT

351.34408 K
= 78.194 °C

12 Heat lost by the fin

Second **Point Evaluation** · select **Point 1** (the base) · clear the Unit field



EXPRESSION

```
-237*pi*0.005^2/4*Tx
```

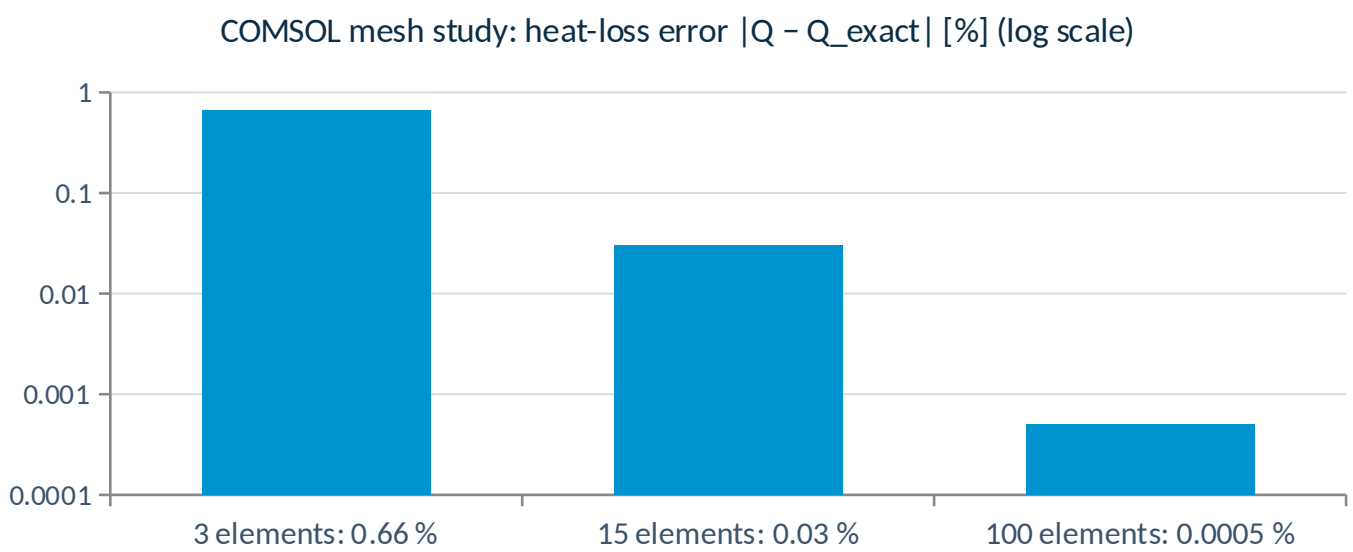
Q = −kA dT/dx at the base; Tx is COMSOL's dT/dx

RESULT

Q = 2.17021 W
(exact 2.17022 W)

COMSOL vs FDM vs exact

	FDM by hand 4 intervals	FDM notebook 10 intervals	COMSOL 100 elements	Exact
Tip temperature [°C]	78.26	78.205	**78.194**	78.194
Heat lost [W]	2.18	2.1721	**2.1702**	2.1702



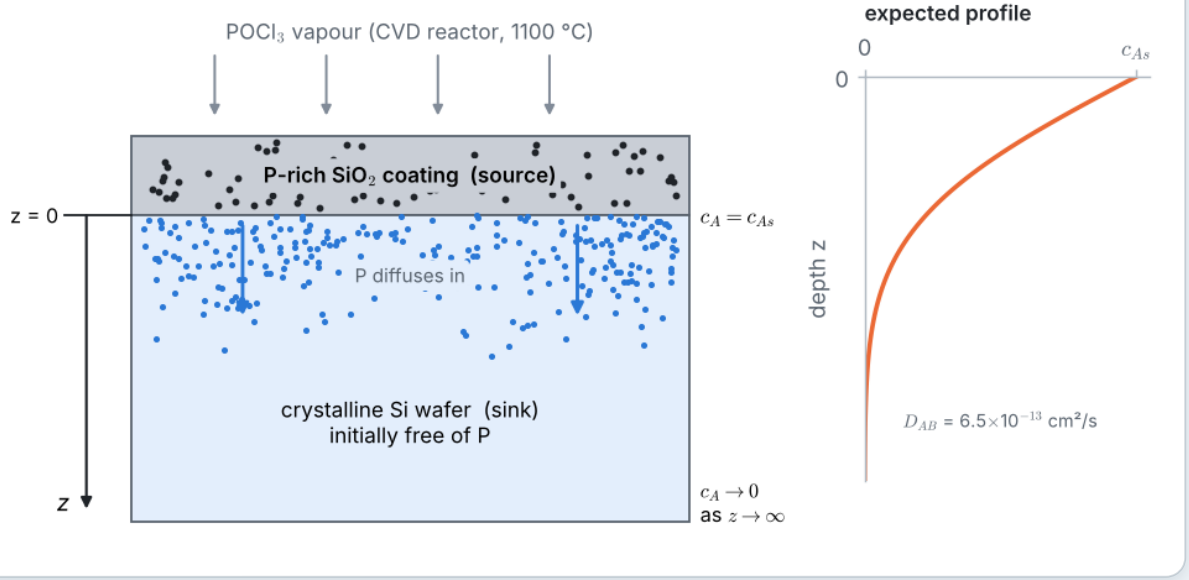
Reading the mesh study

T_{tip} is already exact on the Normal mesh: COMSOL uses quadratic elements. Q needs a **derivative**, so its error falls more slowly, roughly with the element size squared.

TRY IT

Change h to **100 W/(m²·K)** (a fan). **Predict first:** hotter or colder tip? Then check with COMSOL.

Doping a silicon wafer: 1D and unsteady



$c_s = 2.5\times10^{20}$ atoms/cm³

$D = 6.5\times10^{-13}$ cm²/s

1 h at 1100 °C

film edge: c = 1 % of c_s

NEW IN COMSOL — time-dependent · parameters · Diluted Species

GOVERNING EQUATION (FICK'S LAW)

$$\frac{\partial c_A}{\partial t} = D_{AB}\frac{\partial^2 c_A}{\partial z^2}$$

EXACT SOLUTION (SEMI-INFINITE WAFER)

$$\frac{c_A}{c_{As}} = 1 - \operatorname{erf}\left(\frac{z}{2\sqrt{D_{AB}\,t}}\right)$$

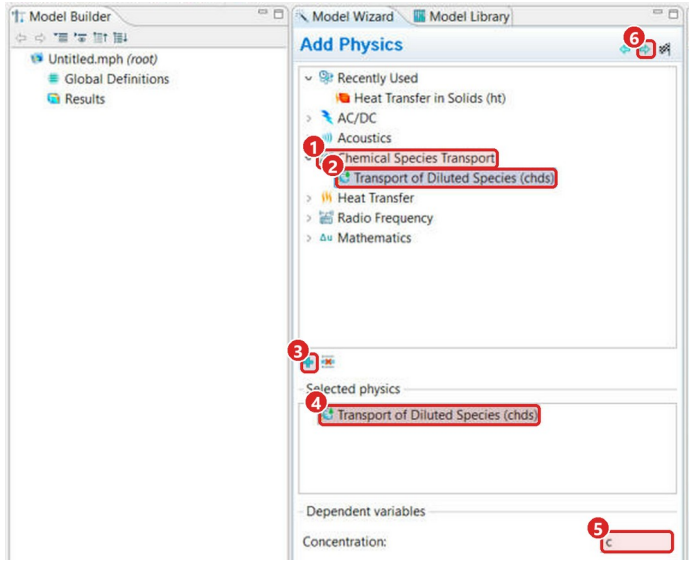
TARGET

z_{film} = 1.762 µm

depth where c = 1 % of c_s, after 1 h (exact)

Model Wizard and parameters

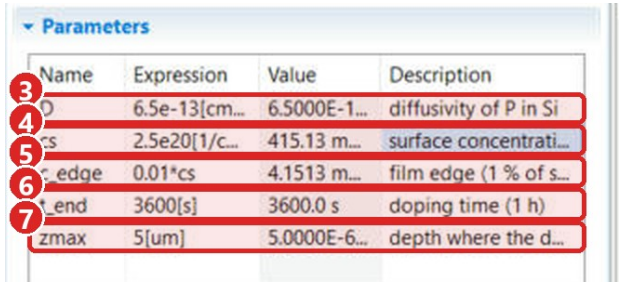
1. **1D** (COMSOL's x is the depth z)
2. Chemical Species Transport › **Transport of Diluted Species**
3. **Time Dependent** › Finish



COMSOL's concentration c is in **mol/m³**: atoms/cm³ must be converted with Avogadro's number.

5. **Parameters: all data in one place**

Name	Expression	Meaning
D	6.5e-13[cm^2/s]	diffusivity of P in Si
cs	2.5e20[1/cm^3]/ 6.02214e23[1/mol]	surface conc., 415.13 mol/m³
c_edge	0.01*cs	film edge (1 %)
t_end	3600[s]	doping time, 1 h
zmax	5[um]	depth of the model



Units in [brackets]: COMSOL converts everything to SI. Change a value and re-run.

Domain, boundary conditions and mesh

Point 1 · z = 0

$c = c_s$

coating: source

Initial value: c = 0 everywhere (the wafer starts free of P)



Point 2 · z = 5 µm

c = 0

deep wafer: sink

$D = 6.5\times10^{-13}$ cm²/s, velocity u = 0: diffusion only

mesh: 50 elements of 0.1 µm

6 Geometry

Interval 0 → **zmax**

7 Diffusion

D_c = **D**, u = 0

8 Surface

Concentration at Point 1: **c = cs**

9 Deep end, start

c = 0 at Point 2; initial c = 0

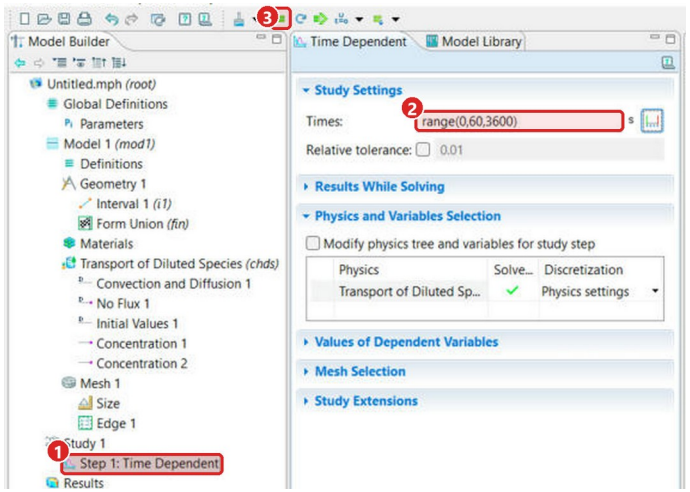
10 Mesh

User-controlled, max **0.1[um]** → 50 elements

Why cut at 5 µm? There $c/c_s = \mathrm{erfc}(5.17) \approx 4\times10^{-13}$: the semi-infinite wafer is effectively infinite.

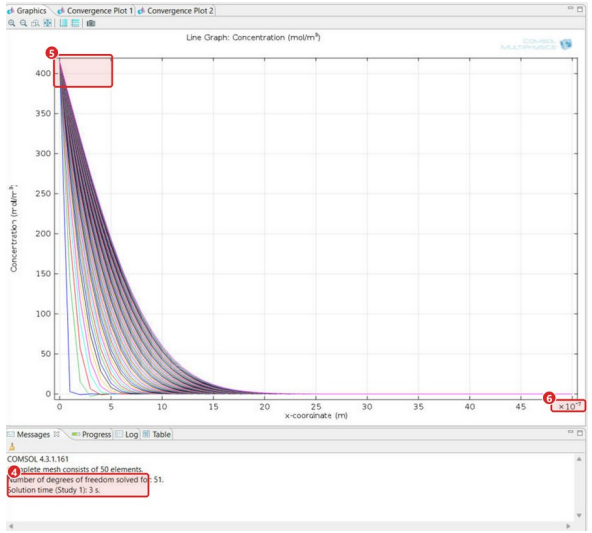
Solve in time, then compare with the exact curve

11 Output times, then Compute



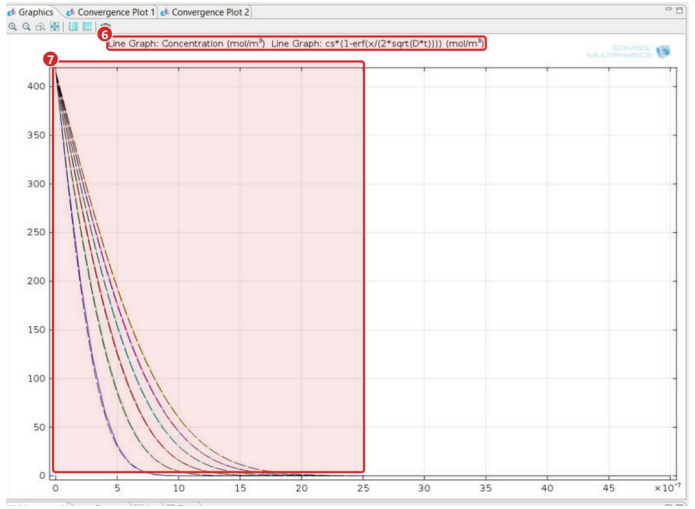
`range(0,60,3600)`

12 Plot every 10 minutes



`range(600,600,3600)`

13 Overlay the exact curve



`cs*(1-erf(x/(2*sqrt(D*t))))`

COMSOL's time stepper is **implicit**: no stability limit on Δt (51 unknowns, solved in about 3 s).

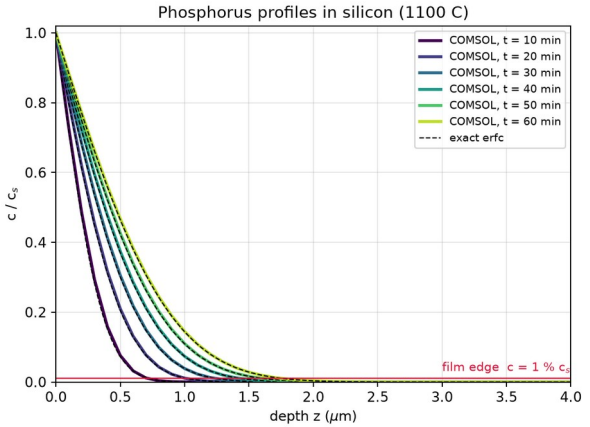
Film depth after 1 hour

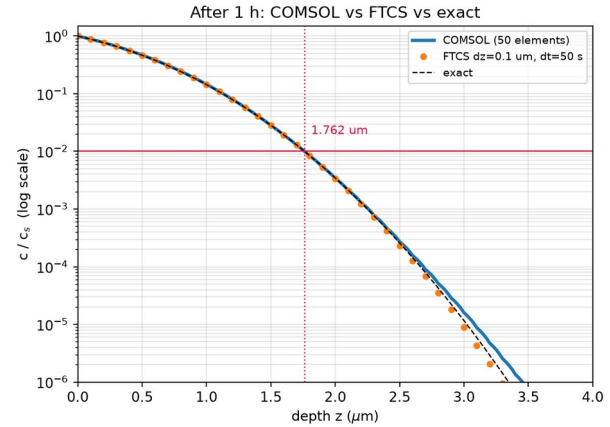
14 Film depth by line integration

Derived Values › Integration › **Line Integration** · Time: **Last** · Unit: **um**

`(c>c_edge)` = 1 where c > 1 % of c_s, so the integral is the film depth

Method	Depth [µm]	Error
Exact (erf solution)	1.7621	–
COMSOL, line integral	1.7722	+0.6 %
COMSOL, exported profile	1.7624	+0.015 %
FTCS, Δz = 0.1 µm, Δt = 50 s	1.7643	+0.12 %





LINK TO PART 1: FTCS STABILITY

$$\Delta t \le \frac{\Delta z^2}{2D} = 76.9 \text{ s}$$

- Δt = 50 s (Fo = 0.33): stable, depth 1.764 µm
- Δt = 100 s (Fo = 0.65): blows up, 10^5 × the surface value!

COMSOL's implicit time stepper has no such limit.

A plate with four interior nodes (2D)

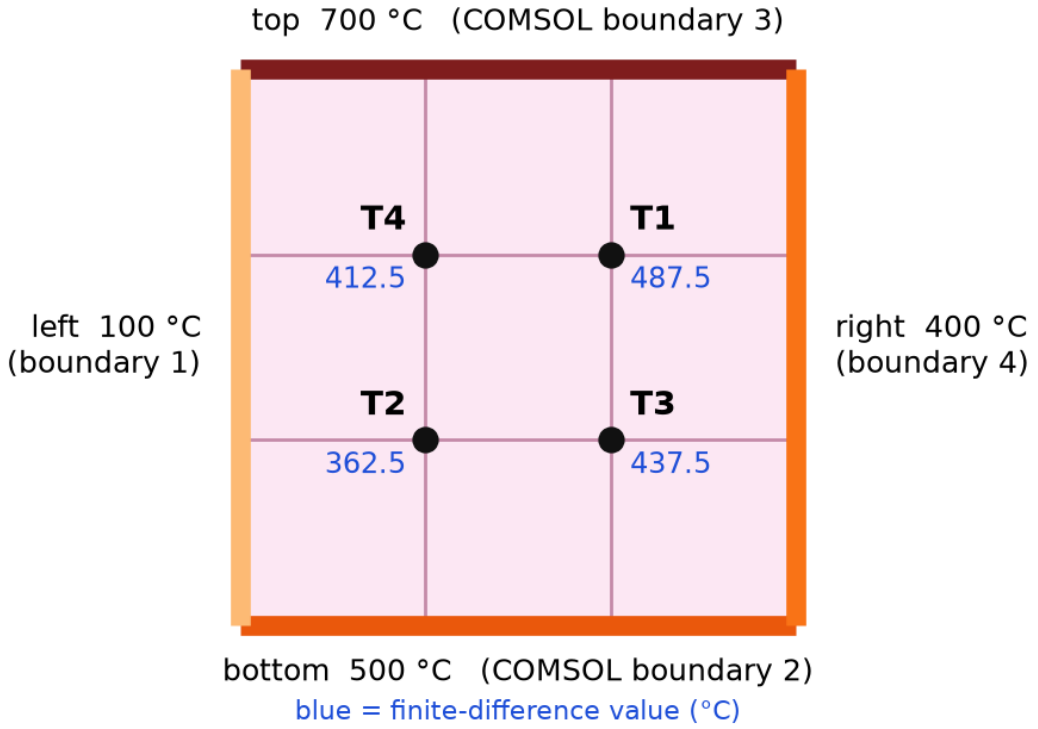


Plate size and k are not given: they are not needed!

LAPLACE EQUATION (STEADY 2D CONDUCTION)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

FDM, Δx = Δy: THE FIVE-POINT STENCIL

$$T_{i,j} = \tfrac{1}{4}(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1})$$

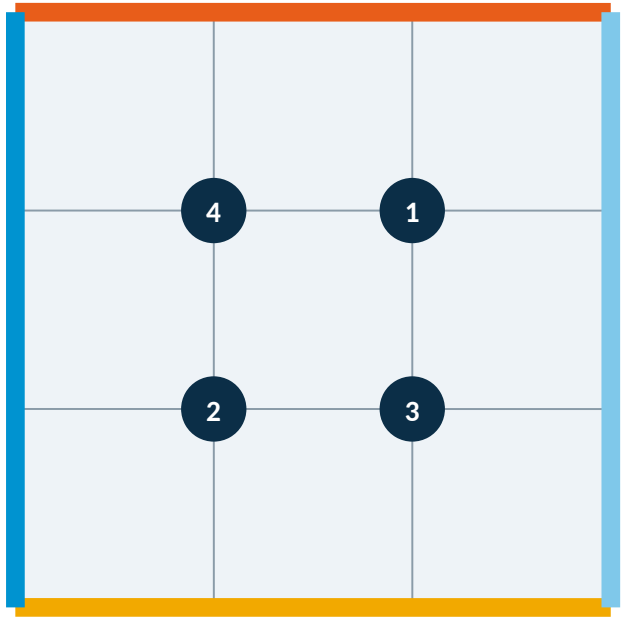
HAND ANSWER (4 UNKNOWNS)

- T_1 = 487.5 °C
- T_2 = 362.5 °C
- T_3 = 437.5 °C
- T_4 = 412.5 °C

NEW — 2D geometry · mesh sweep · heat-balance check

Build the plate: 2D, a square, four edge temperatures

700 °C · boundary 3



100 °C
bdry 1

400 °C
bdry 4

500 °C · boundary 2

unit square, side 1 m, corner at (0, 0)

1–3 **Model Wizard**

2D › Heat Transfer in Solids › **Stationary**

5 **Geometry**

Right-click Geometry 1 › **Square**, side 1 m › Build

6 **Material**

$k = 1$, $\rho = 1$, $C_p = 1$: size and k drop out of the Laplace equation

7 **Top edge**

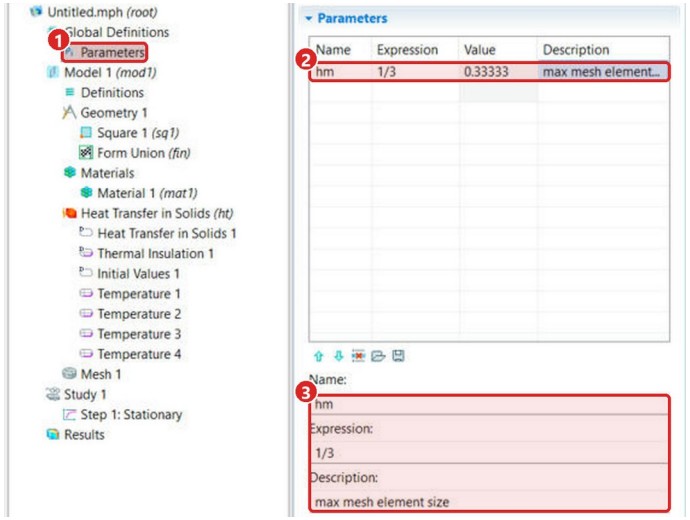
Temperature on boundary 3: T_0 = **700[degC]**

8 **Other edges**

boundary 1 → 100[degC] · 4 → 400[degC] · 2 → 500[degC]

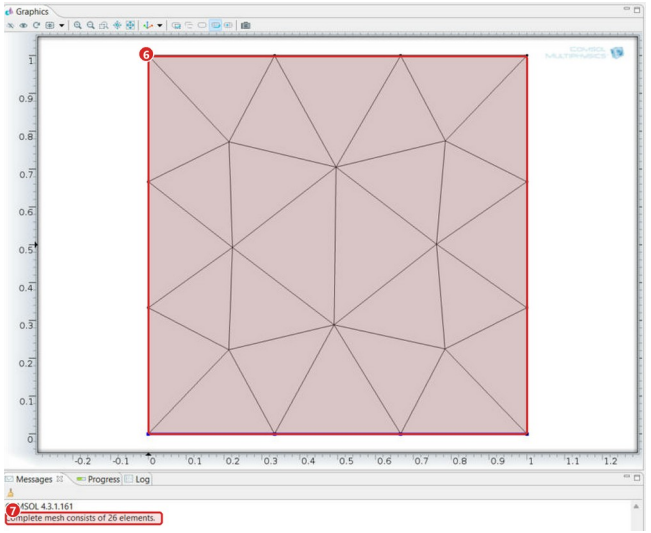
Mesh size as a parameter: a grid study in one click

9 Parameter hm = 1/3



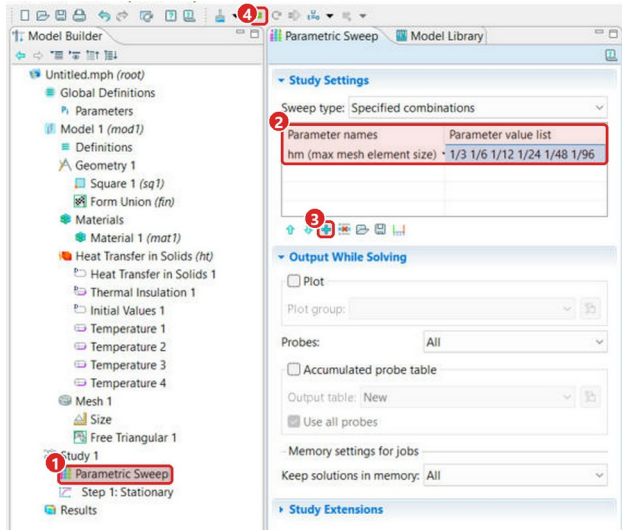

```
hm = 1/3
```

10 Mesh size = hm




```
Maximum element size: hm
```

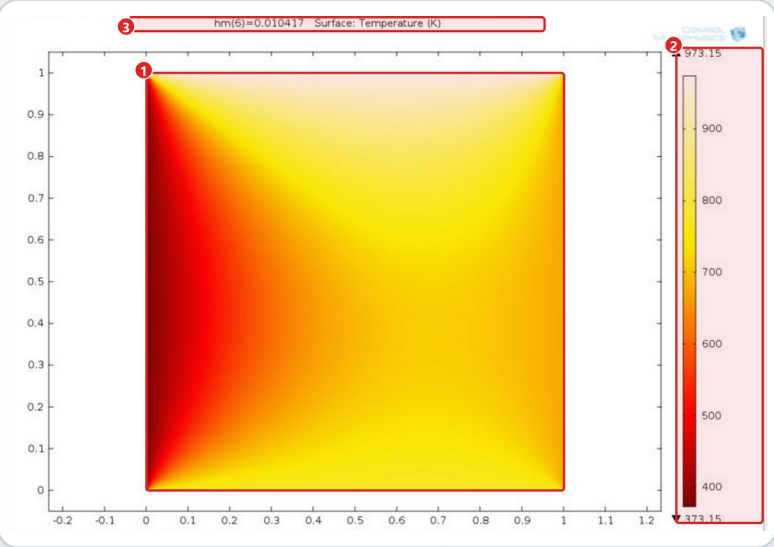
11 Parametric sweep over hm

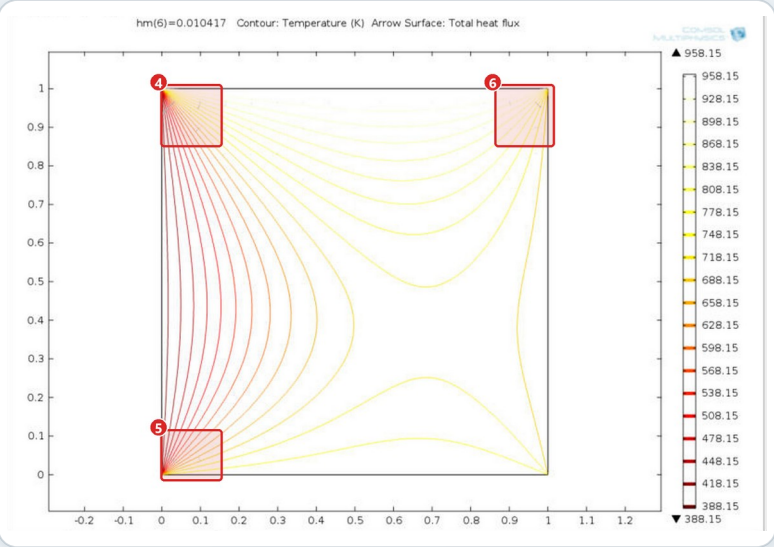



```
1/3 1/6 1/12 1/24 1/48 1/96
```

One Compute solves **6 meshes**, from 26 elements upwards: a grid-convergence study, exactly the verification of Pillar 7.

Results: field, points and heat balance





hm	Total normal heat flux (W/m)
0.33333	-13.41023
0.16667	1.19722
0.08333	1.63032
0.04167	0.99141
0.02083	1.85953
0.01042	2.31662

Step 15: total heat flow through all four edges, per mesh. It shrinks towards 0 as the mesh is refined, apart from the singular corners: heat in = heat out.

13 Cut Point 2D

the 4 points (1/3, 1/3), (2/3, 2/3), (2/3, 1/3), (1/3, 2/3)

14 Point Evaluation

expression T, unit degC, all hm: one row per mesh

15 Heat balance

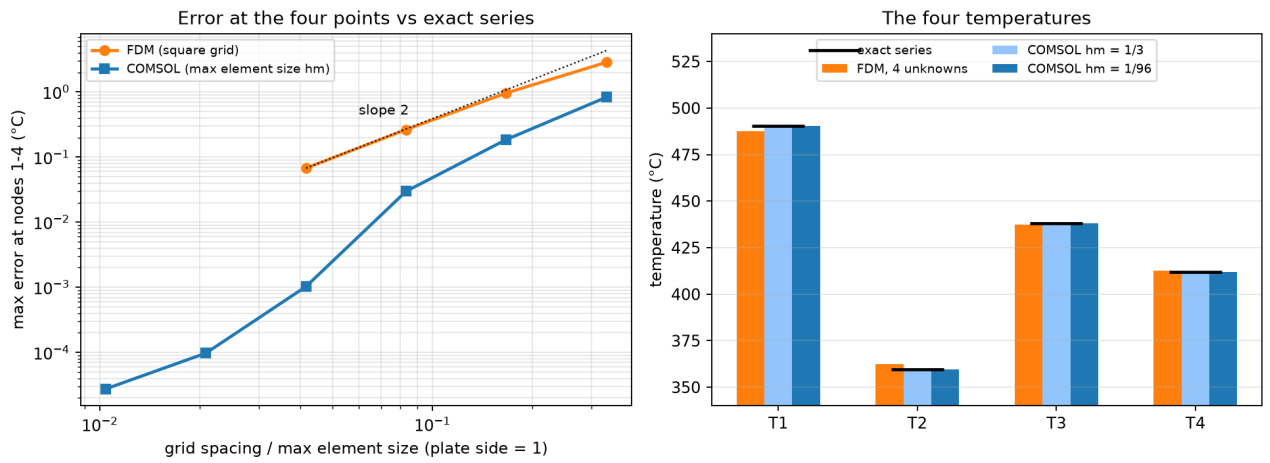
Line Integration of **ht.ntflux** over all edges

16 Isotherms

crowd into the corners where very different edge temperatures meet

Four unknowns, already within 3 °C

	T_1	T_2	T_3	T_4 [°C]
FDM, 4 unknowns	487.5	362.5	437.5	412.5
COMSOL, coarse (hm = 1/3)	490.92	358.81	438.45	411.20
COMSOL, fine (hm = 1/96)	**490.36**	**359.64**	**438.07**	**411.93**
Exact series solution	**490.36**	**359.64**	**438.07**	**411.93**

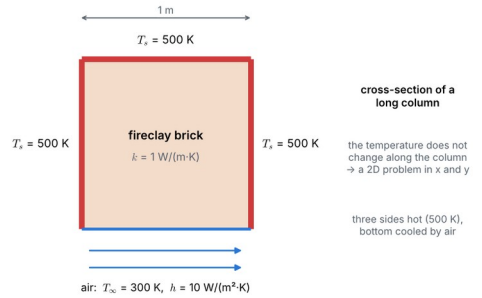


2.9 °C
largest error of the 4-unknown FDM

0.0001 °C
COMSOL fine mesh vs the exact series

slope 2
refining the FDM grid: second-order convergence

More problems in the repository

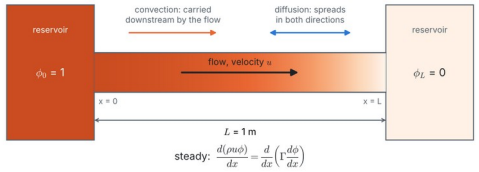


4 · Brick column

2D · Robin BC · symmetry

A coarse 0.25 m grid gets the temperatures right but over-predicts the heat loss to the air by 41 %.

COMSOL guide

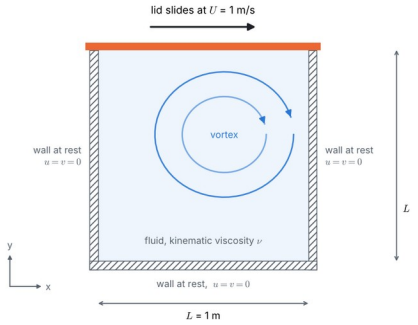


5 · Convection–diffusion

1D · Péclet number

Cell Péclet > 2: central differences wiggle. COMSOL's stabilisation acts like upwinding.

COMSOL guide

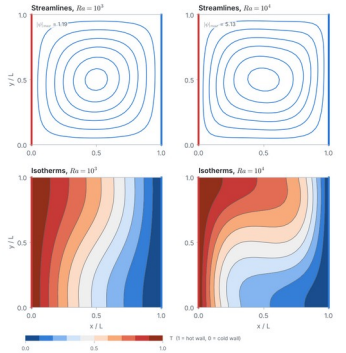


6 · Lid-driven cavity

2D · Navier–Stokes (ψ–ω)

The classic benchmark: the FDM solution converges to Ghia et al. (1982).

COMSOL guide



7 · Natural convection

2D · buoyancy · Nusselt

Heat drives the flow and the flow carries the heat. Richardson extrapolation.

notebook + whiteboard

Key takeaways

Every CFD analysis rests on 7 pillars

Pre-processing (geometry, grid, physics) → solving (scheme, setup, compute) → post-processing (visualize & verify).

FDM turns derivatives into differences

The Taylor series sets the order of accuracy: halve Δx and a 2nd-order error falls by 4.

Check before you trust

Stability limits ($r \le \frac{1}{2}$, CFL), grid refinement, exact solutions and benchmarks.

COMSOL automates the pillars, not the thinking

You still choose the physics, boundary conditions and mesh, and you verify the answer.

AIChE
KAUST Student Chapter

Thank you!

Questions? Let's discuss.

FURTHER READING

- Anderson: CFD, The Basics with Applications (1995)
- Versteeg & Malalasekera: An Introduction to CFD, the FVM (2007)
- Patankar: Numerical Heat Transfer and Fluid Flow (1980)
- Barba & Forsyth: CFD Python, 12 steps to Navier–Stokes (free)

simple_CFD repository

github.com/maryamomer2004/simple_CFD